Relatório de Turnover

Análise Exploratória + Importância de Variáveis

Arquivo analisado: IBM-HR-Employee-Attrition.csv

Coluna de Turnover: Attrition

Data de geração: 18/03/2026 13:27

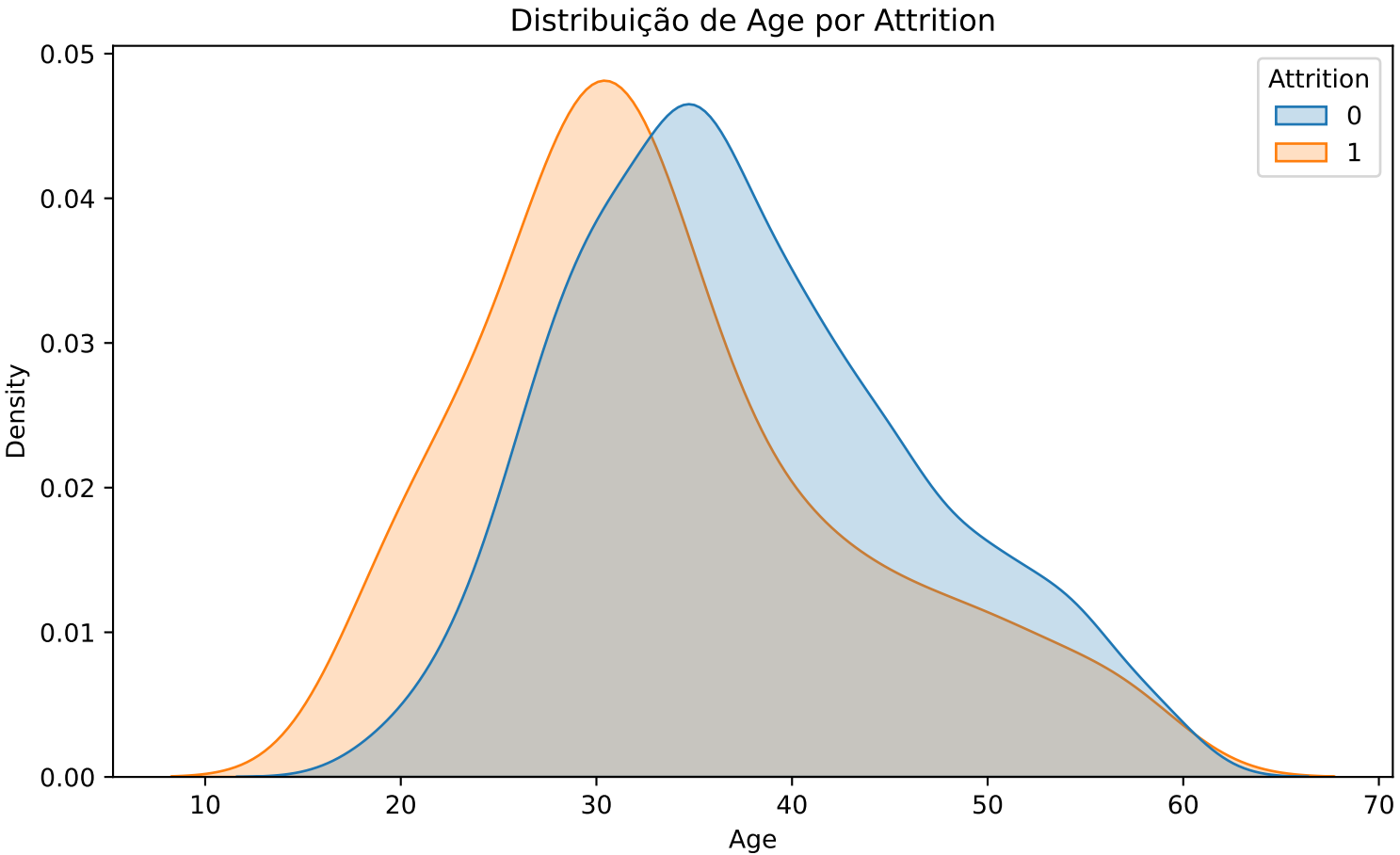
Resumo interpretativo para RH

Relatório: IBM-HR-Employee-Attrition
Leitura em linguagem simples para apoio à área de Recursos Humanos

1) Resumo executivo
Desenvolvemos um modelo para prever se um colaborador ficará ou sairá da empresa, usando dados de 1470 funcionários. O modelo identifica alguns cargos e hábitos de viagem e horas extras como mais ligados à saída. Entretanto, a precisão dessa previsão ainda é moderada, indicando que o modelo pode ajudar, mas não deve ser o único critério para decisões.

2) O que isso sugere
- Funcionários em cargos como Técnico de Laboratório e Representante de Vendas têm mais influência na previsão de saída.
- A prática de fazer ou não horas extras também é um indicador importante relacionado ao turnover.
- A frequência em viagens a trabalho aparece como um fator relevante para entender o comportamento de saída.
- O modelo não é perfeito (F1 de 0.47), ou seja, pode errar bastante ao prever quem vai sair.

3) Recomendações práticas para investigação
- Avaliar as condições e satisfação dos funcionários nos cargos de Técnico de Laboratório, Diretor de Pesquisa e Vendas.
- Analisar o impacto das horas extras no engajamento e desgaste dos colaboradores.
- Investigar como a frequência de viagens a trabalho afeta a experiência e a decisão pela permanência.
- Complementar os resultados do modelo com pesquisas internas, entrevistas ou outras métricas qualitativas para entender melhor os motivos do turnover.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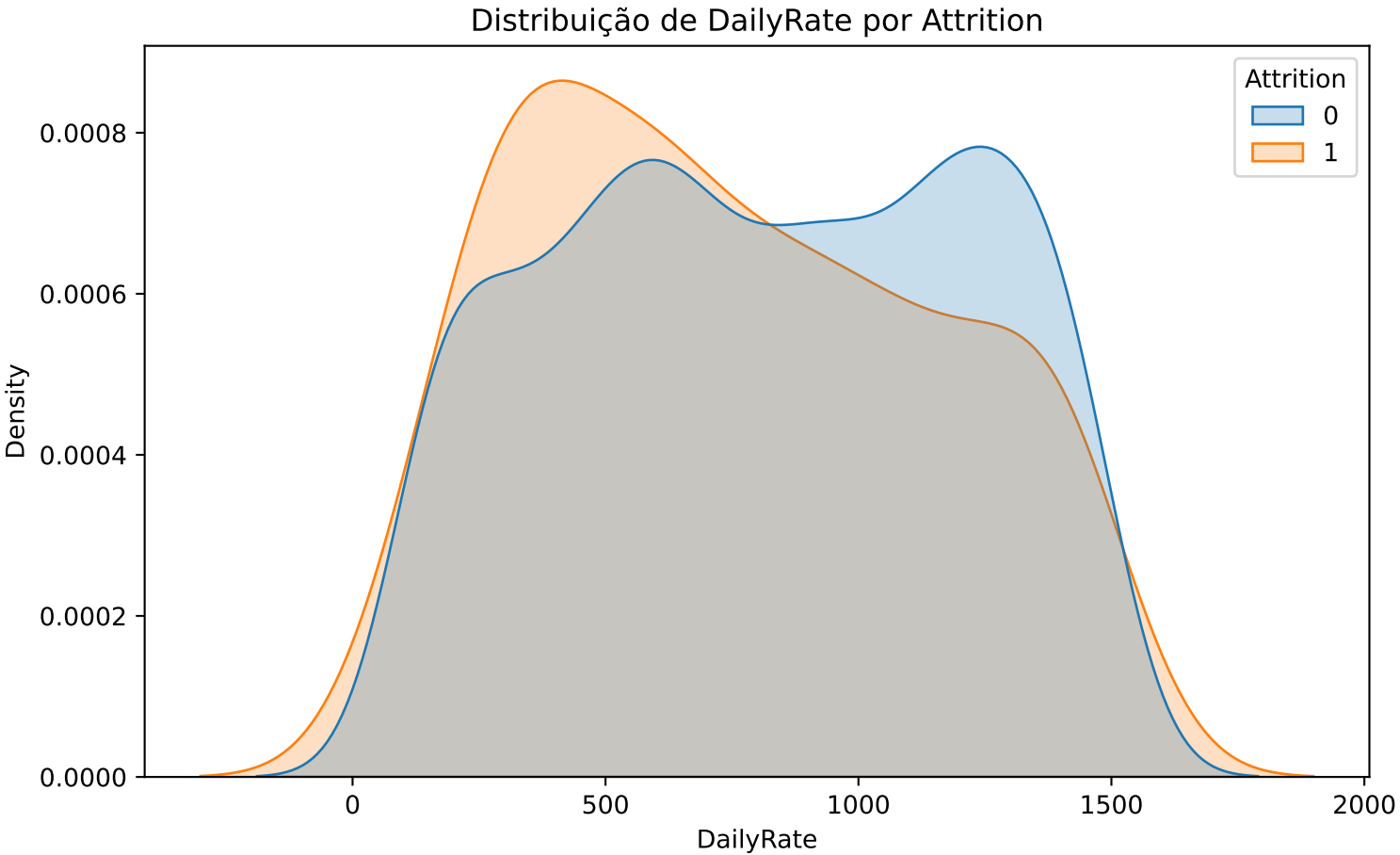
50
60
70
Age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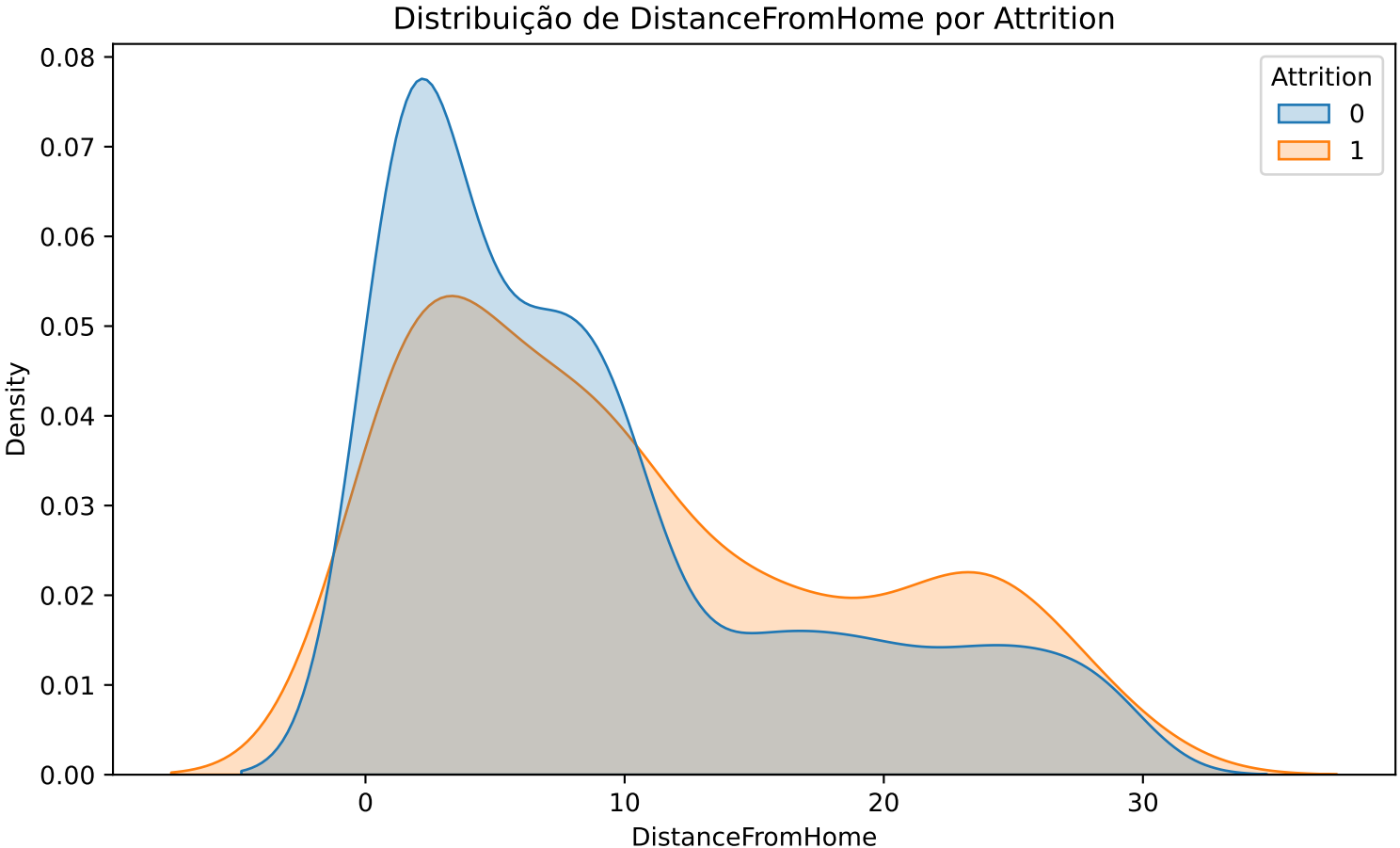
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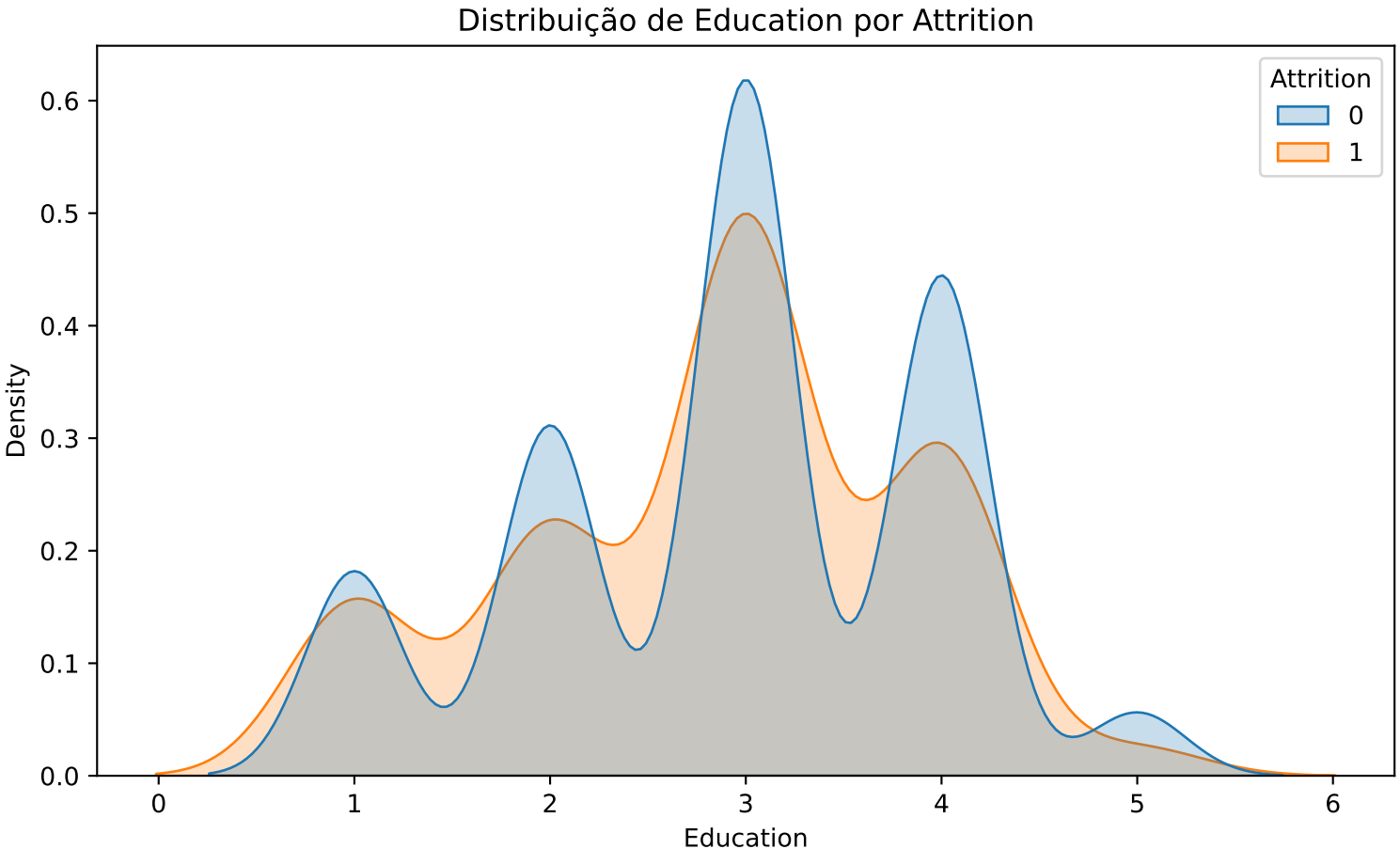
DistanceFromHome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1
2
3
4
5
6
Education

Distribuição de EmployeeCount por Attrition
Attrition
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1
Density
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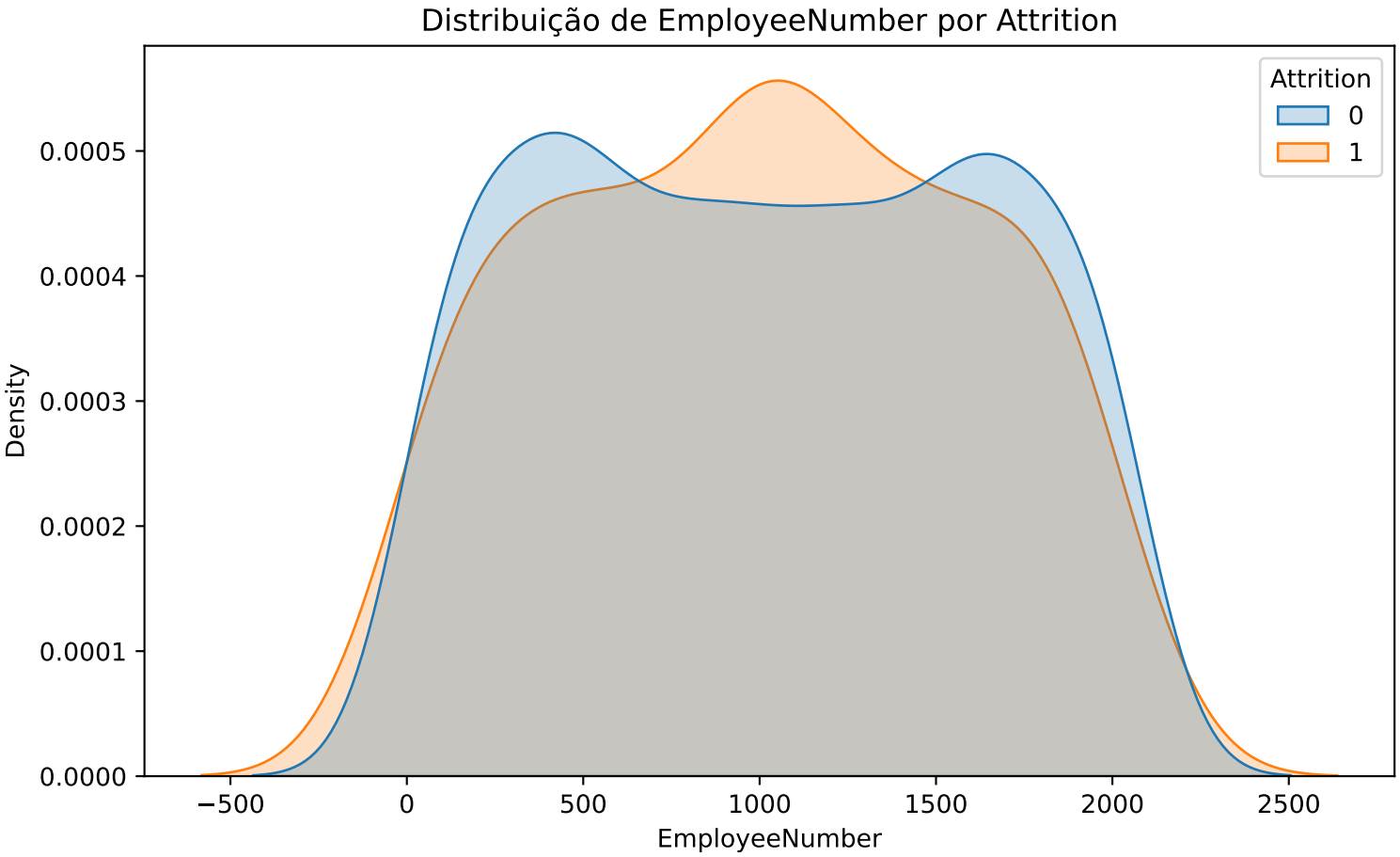
Distribuição de EmployeeNumber por Attrition
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2000
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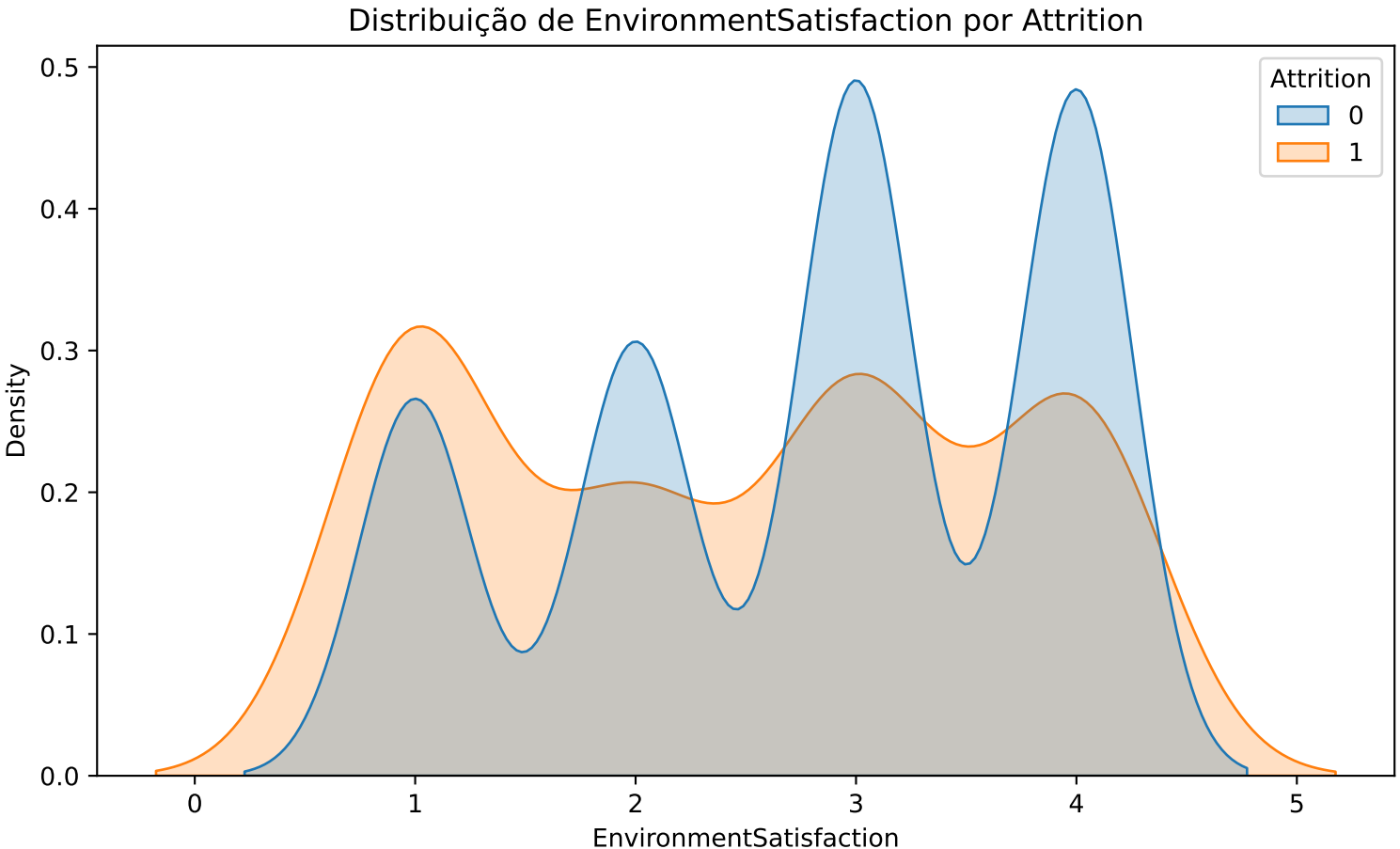
EmployeeNumber


Distribuição de EnvironmentSatisfaction por Attrition
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Density
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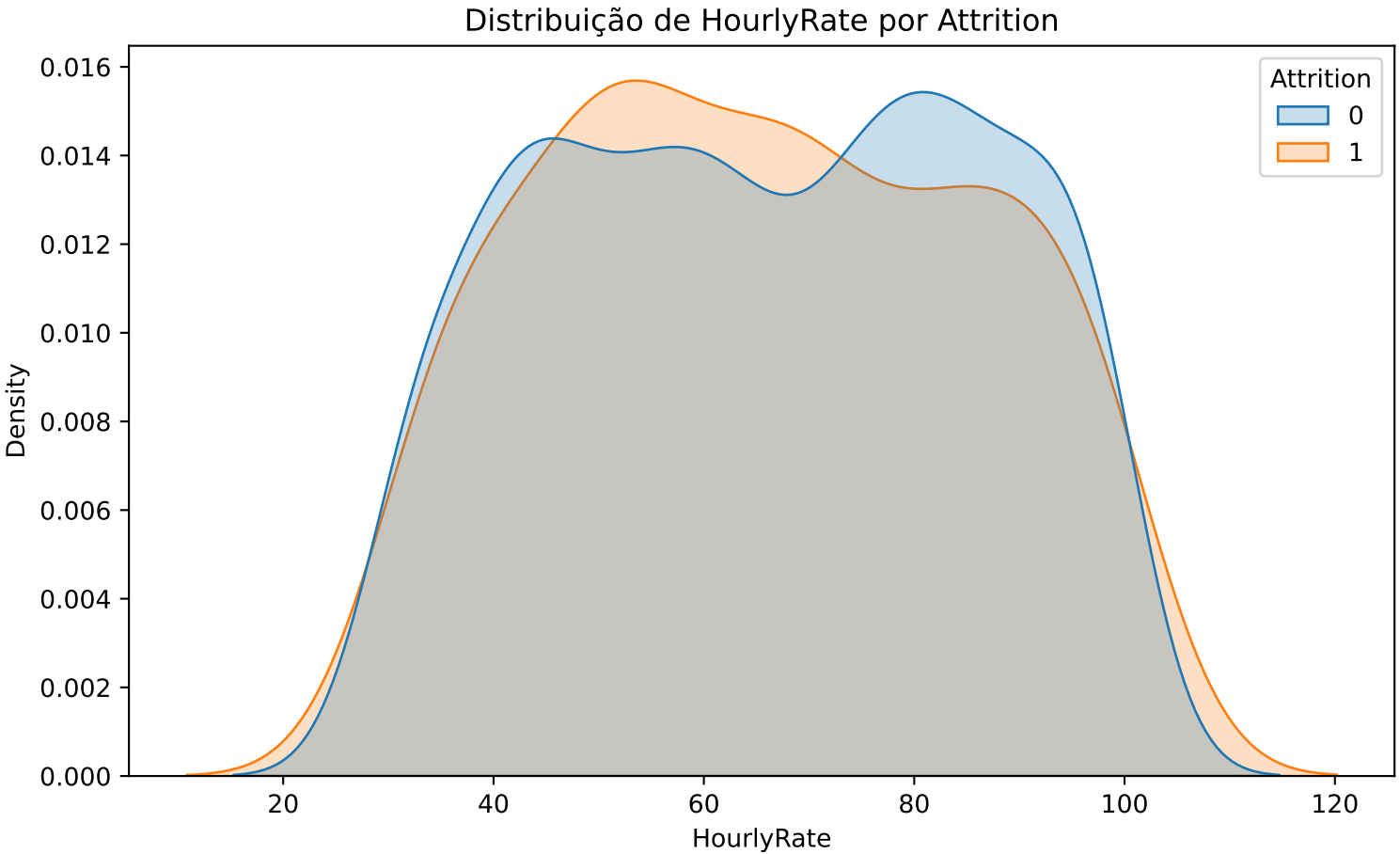
1
2
3
4
5
EnvironmentSatisfaction


Distribuição de HourlyRate por Attrition
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0.002
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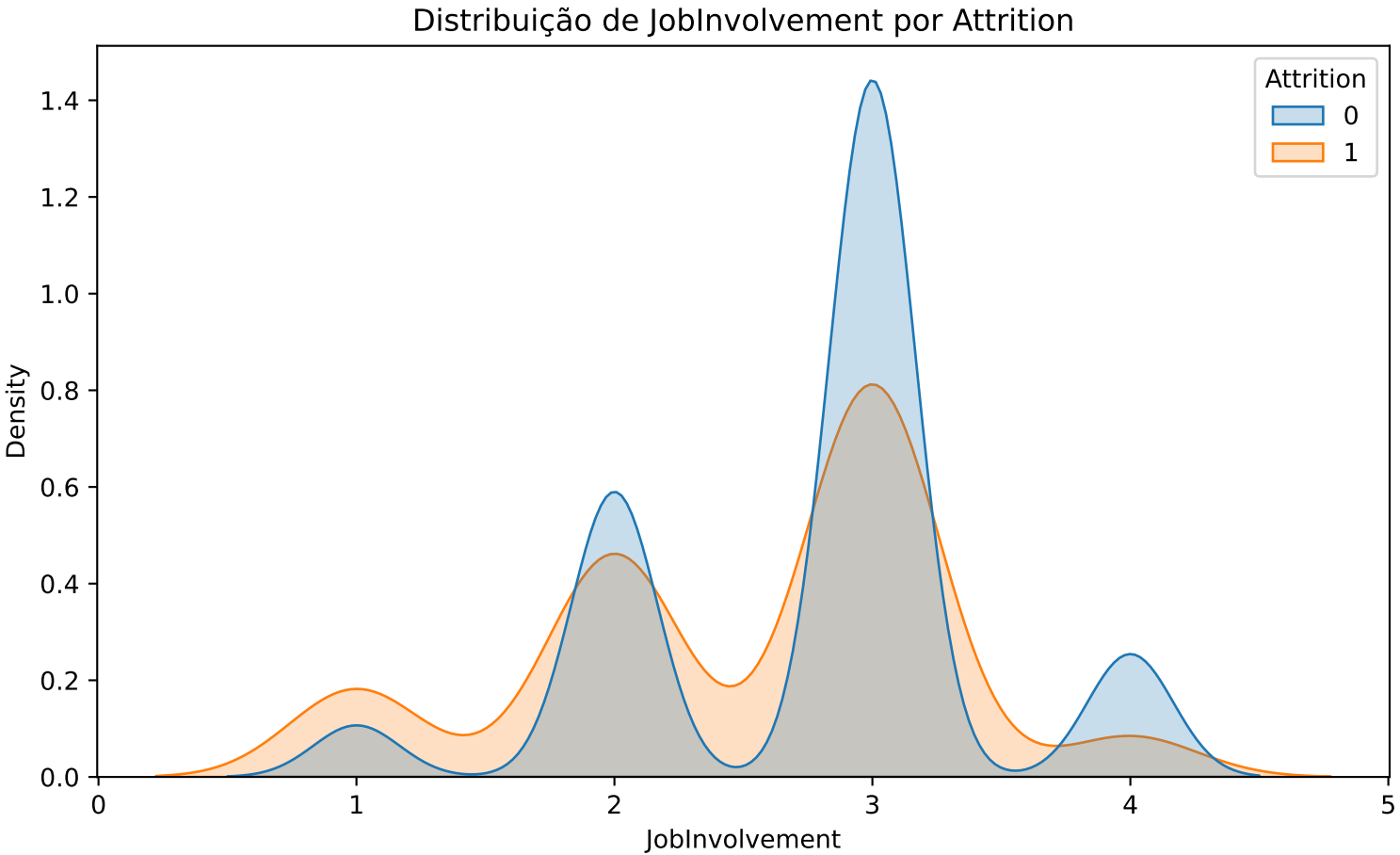
HourlyRate


Distribuição de JobInvolvement por Attrition
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Density
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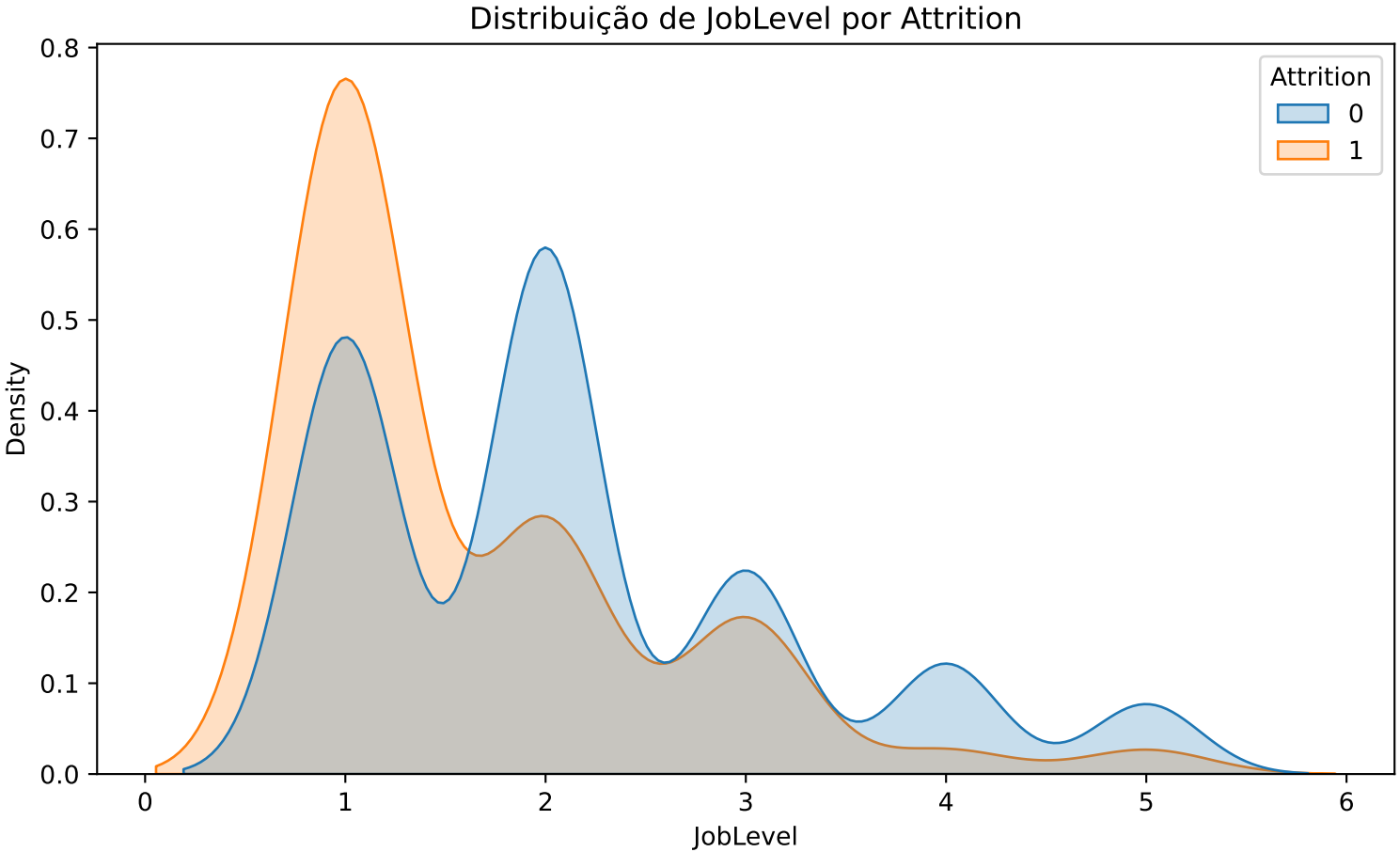
0
1
2
3
4
5
JobInvolvement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Density
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1
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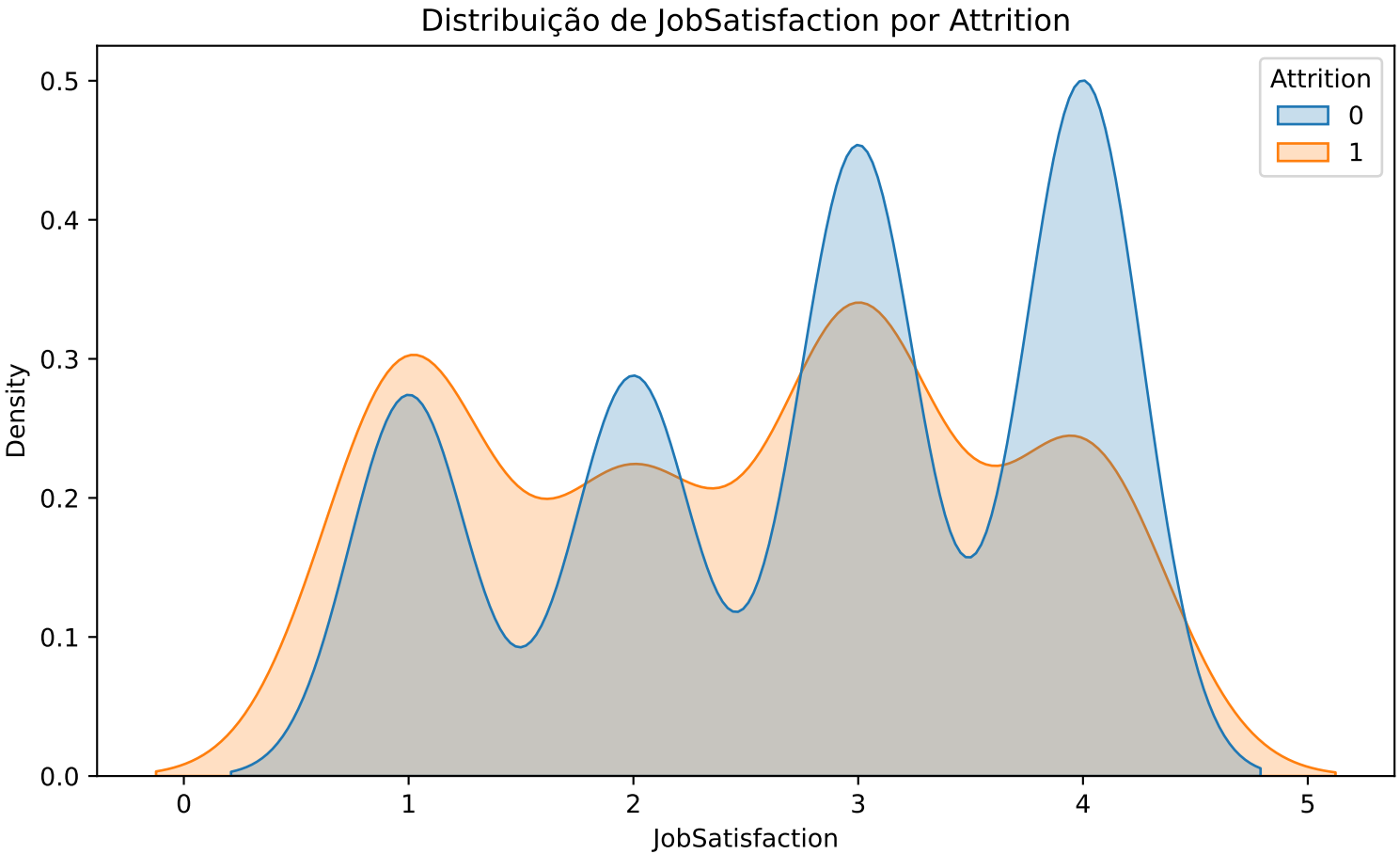
3
4
5
6


Distribuição de JobSatisfaction por Attrition
Attrition
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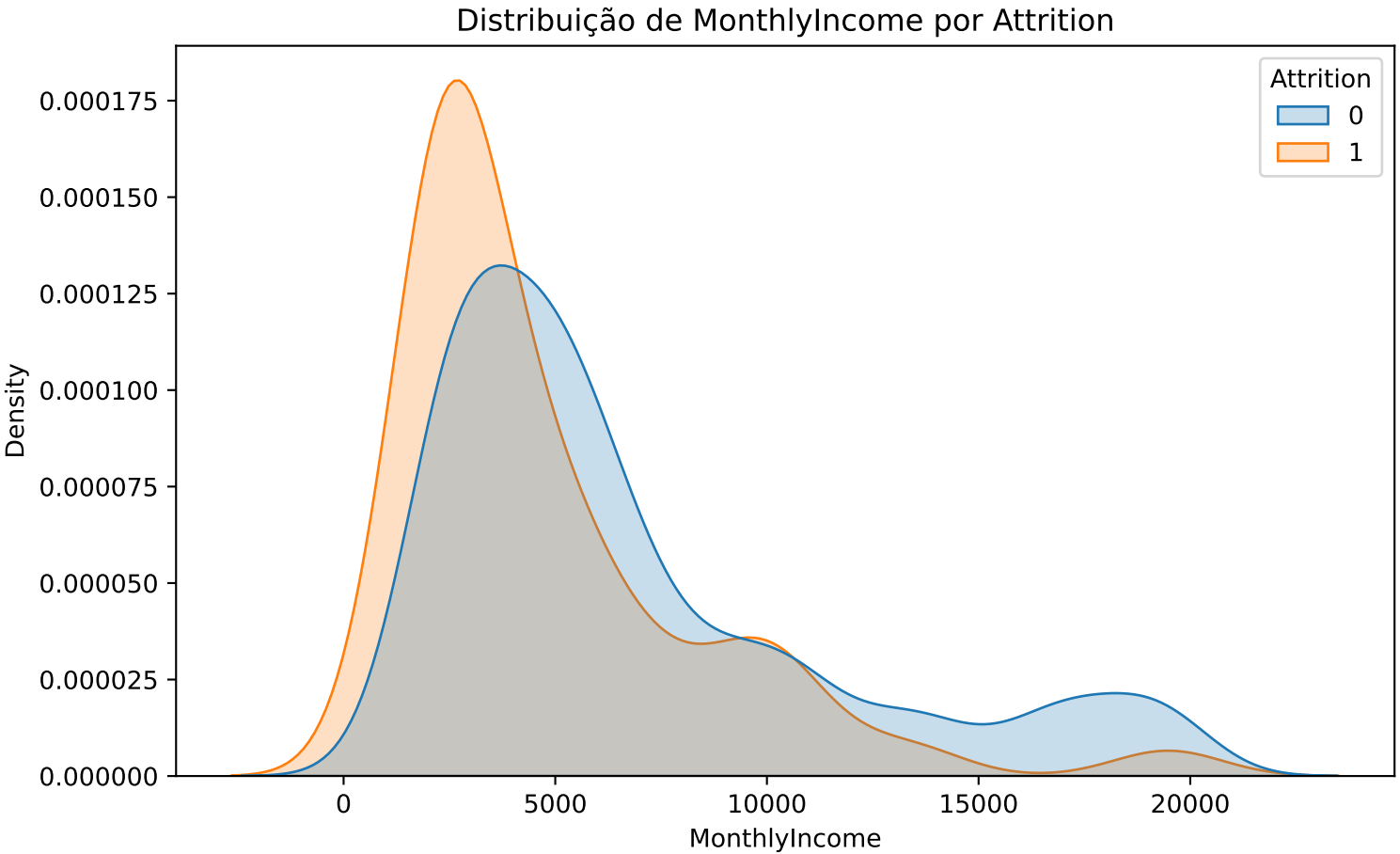
0
1
2
3
4
5
JobSatisfaction


Distribuição de MonthlyIncome por Attrition
Attrition
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1
Density
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0.000125
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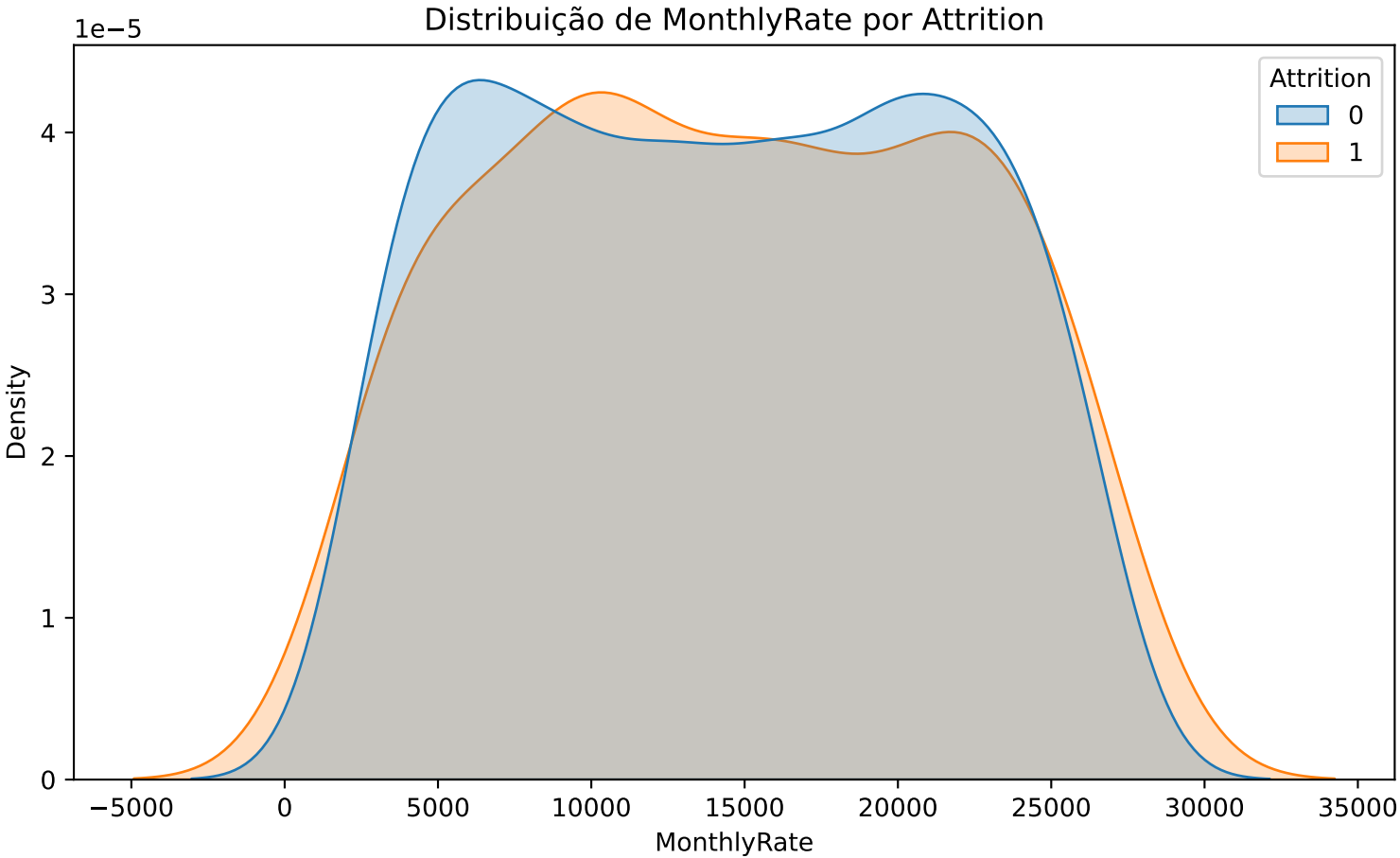
0.000075
0.000050
0.000025
0.000000
0
5000
10000
15000
20000
MonthlyIncome


Distribuição de MonthlyRate por Attrition
1e−5
Density
MonthlyRate
Attrition
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10000
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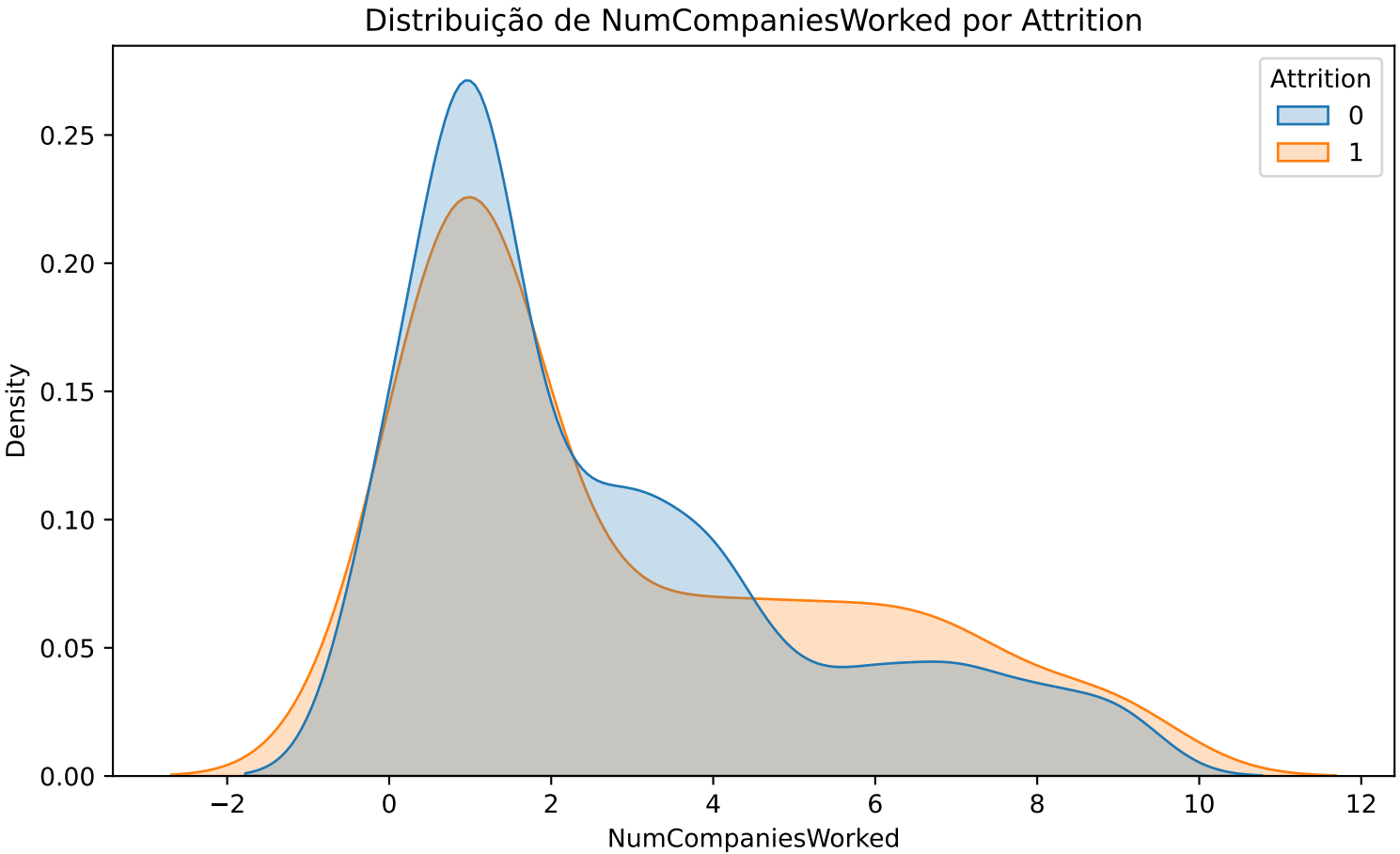
Distribuição de NumCompaniesWorked por Attrition
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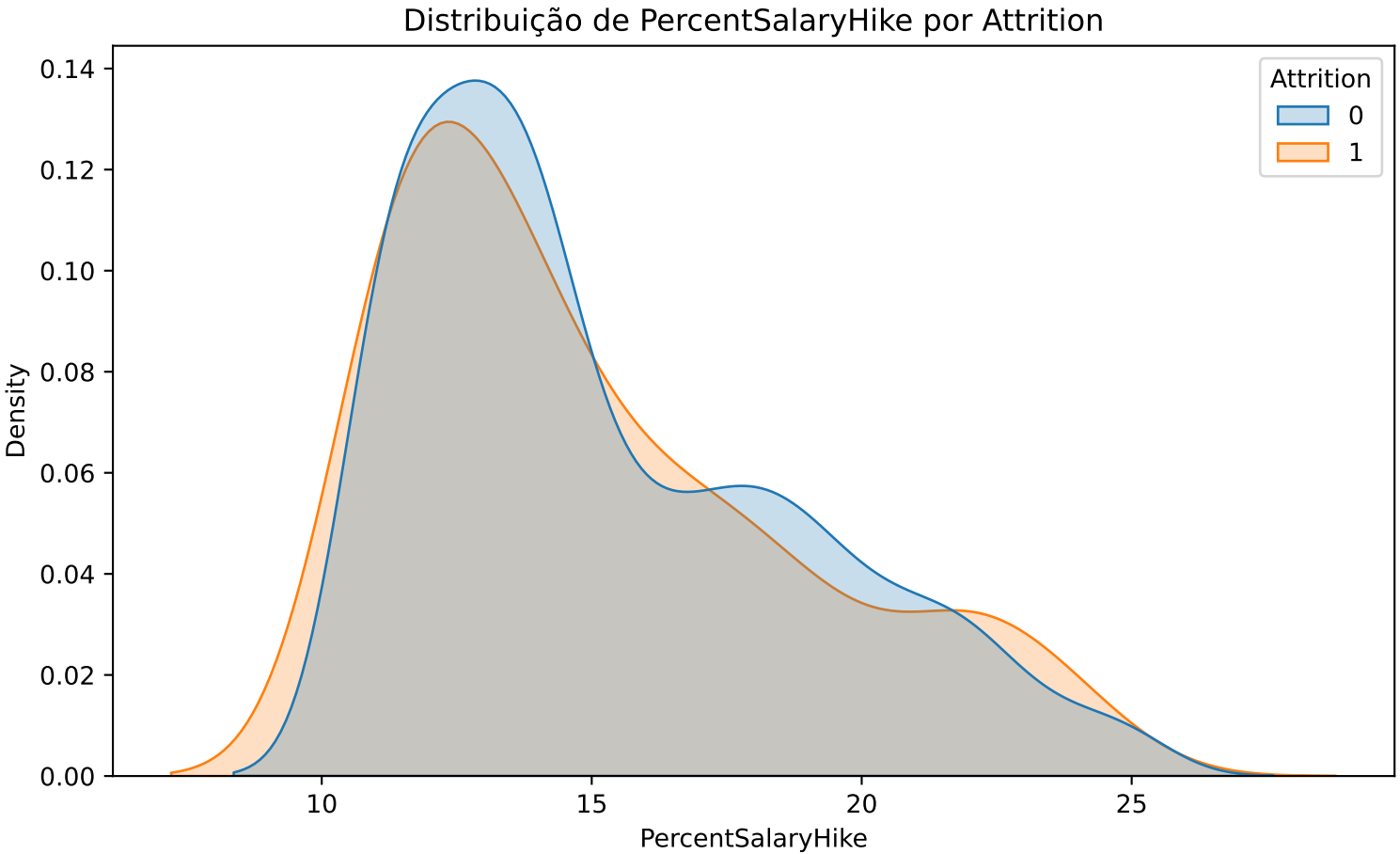
2
4
6
8
10
12
NumCompaniesWorked


Distribuição de PercentSalaryHike por Attrition
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1
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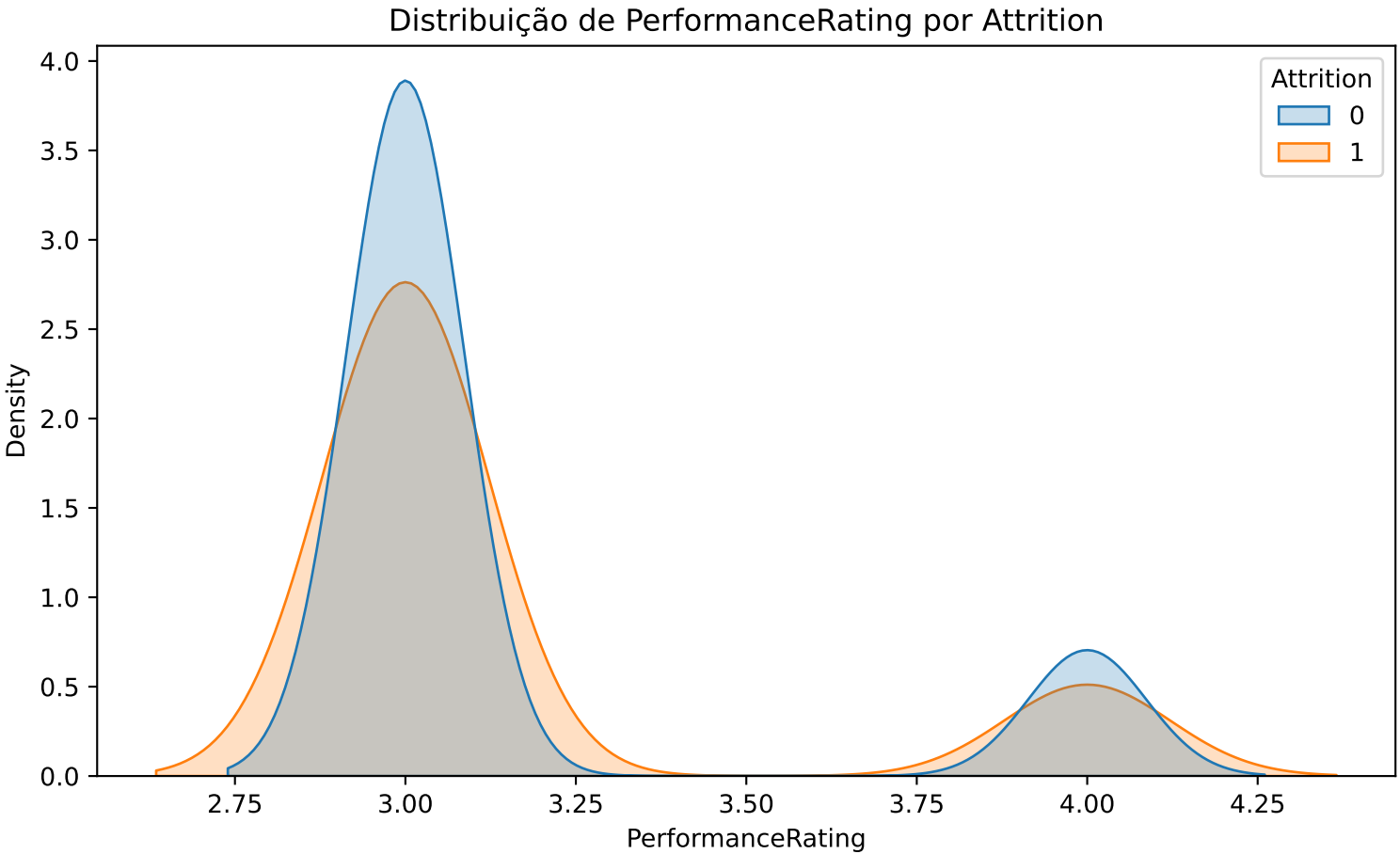
0.00
10
15
20
25
PercentSalaryHike


Distribuição de PerformanceRating por Attrition
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Density
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3.0
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3.75
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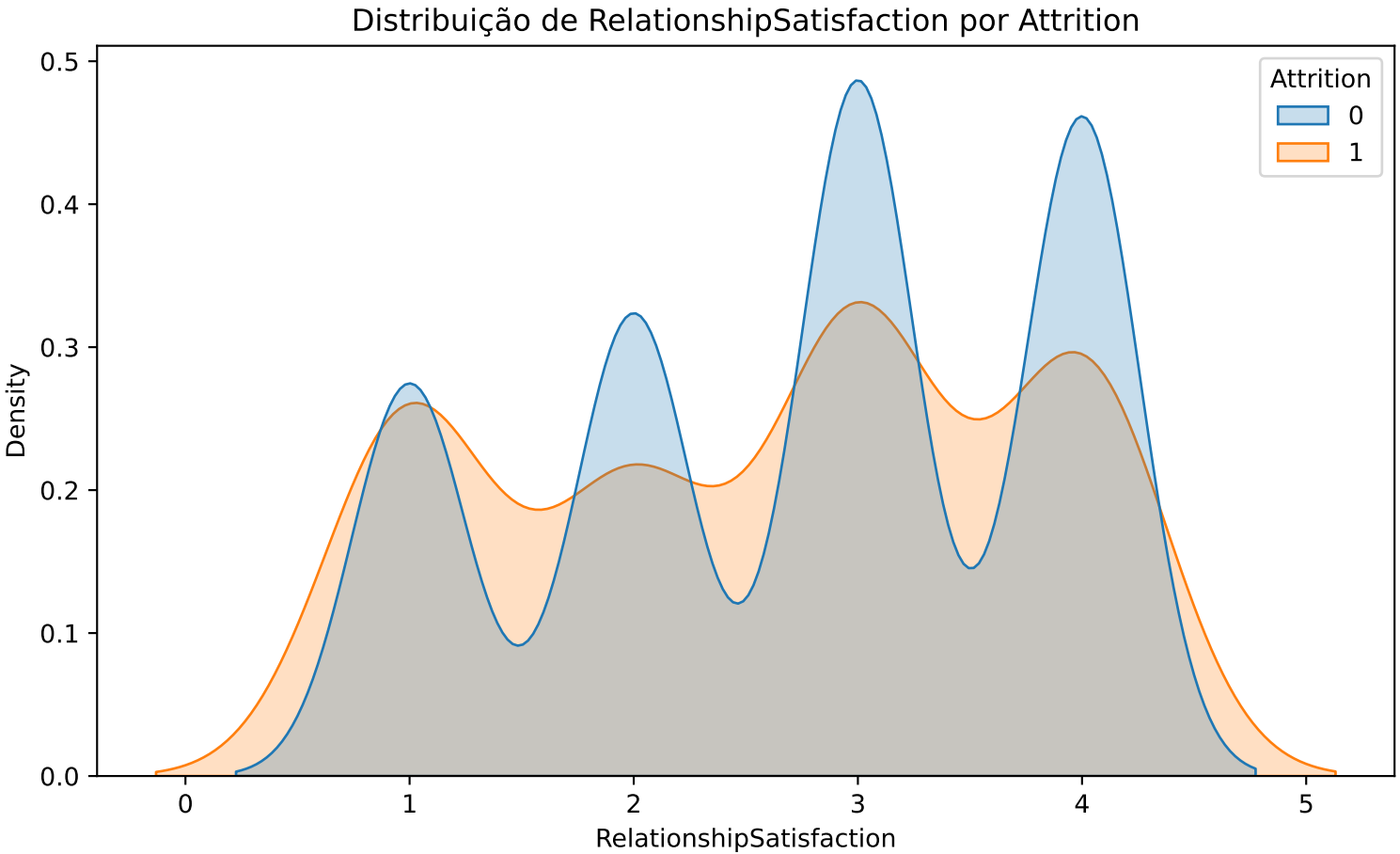
4.25
PerformanceRating


Distribuição de RelationshipSatisfaction por Attrition
Attrition
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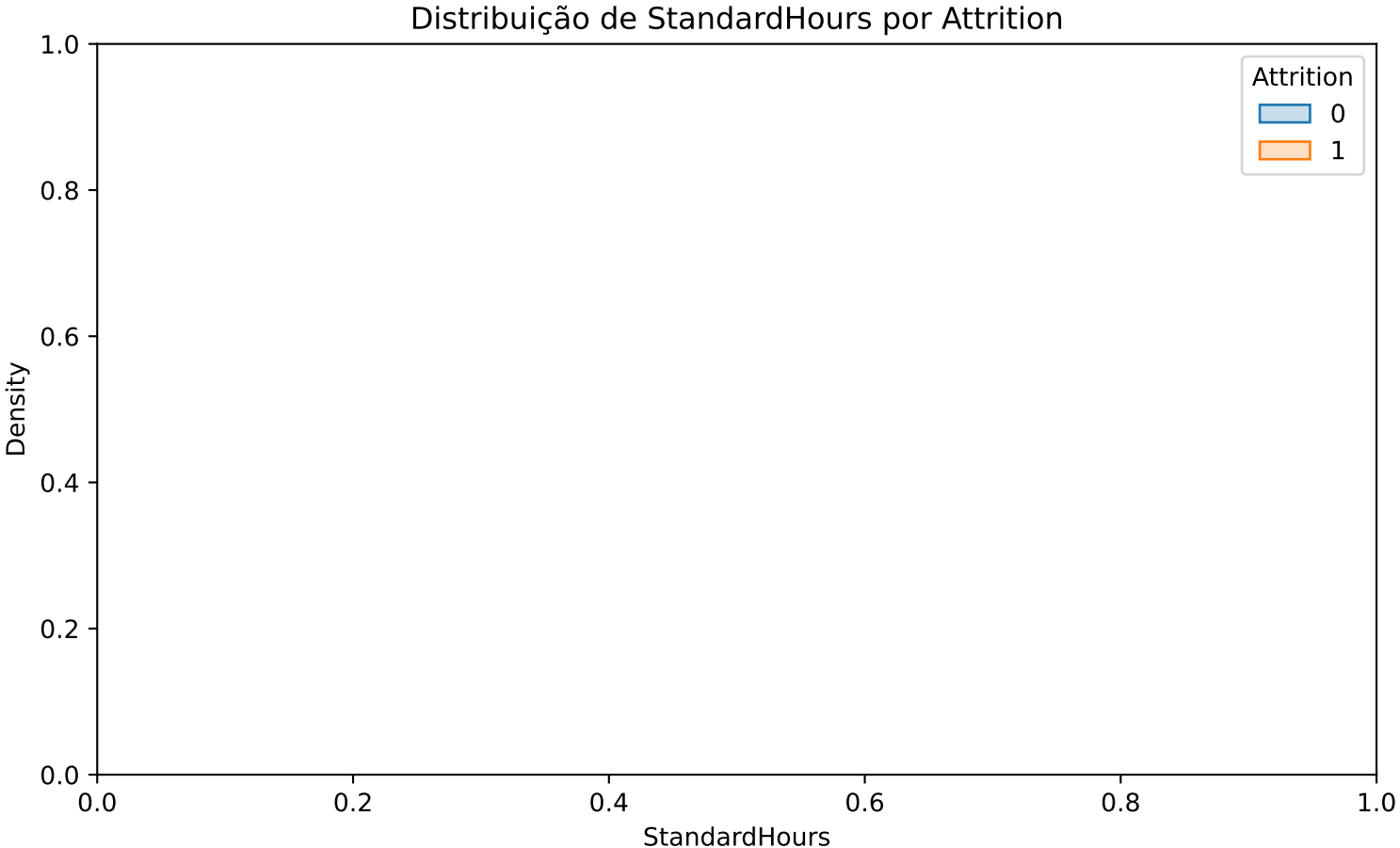
0
1
2
3
4
5
RelationshipSatisfaction


Distribuição de StandardHours por Attrition
Attrition
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Density
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0.8
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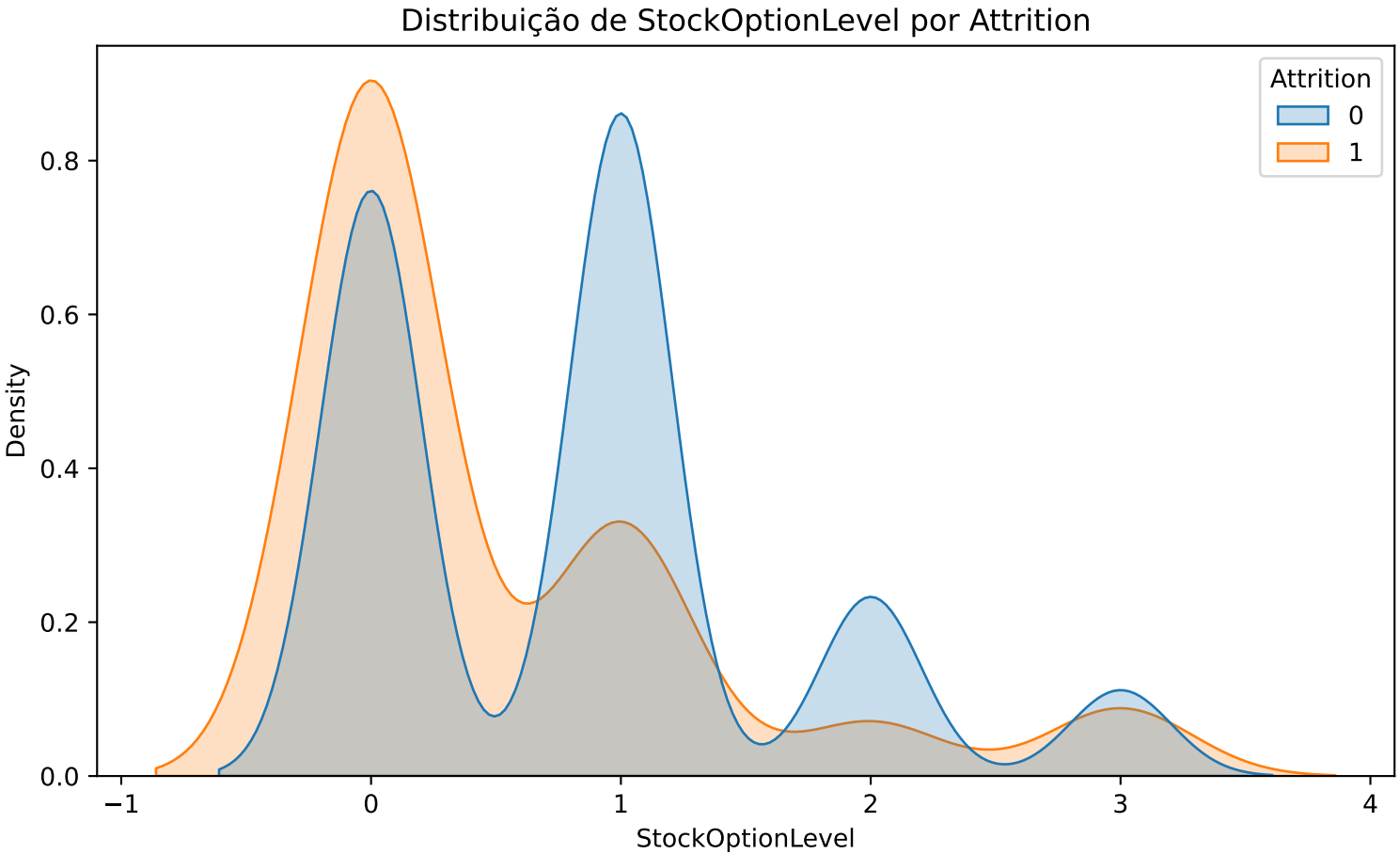
StandardHours


Distribuição de StockOptionLevel por Attrition
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0.6
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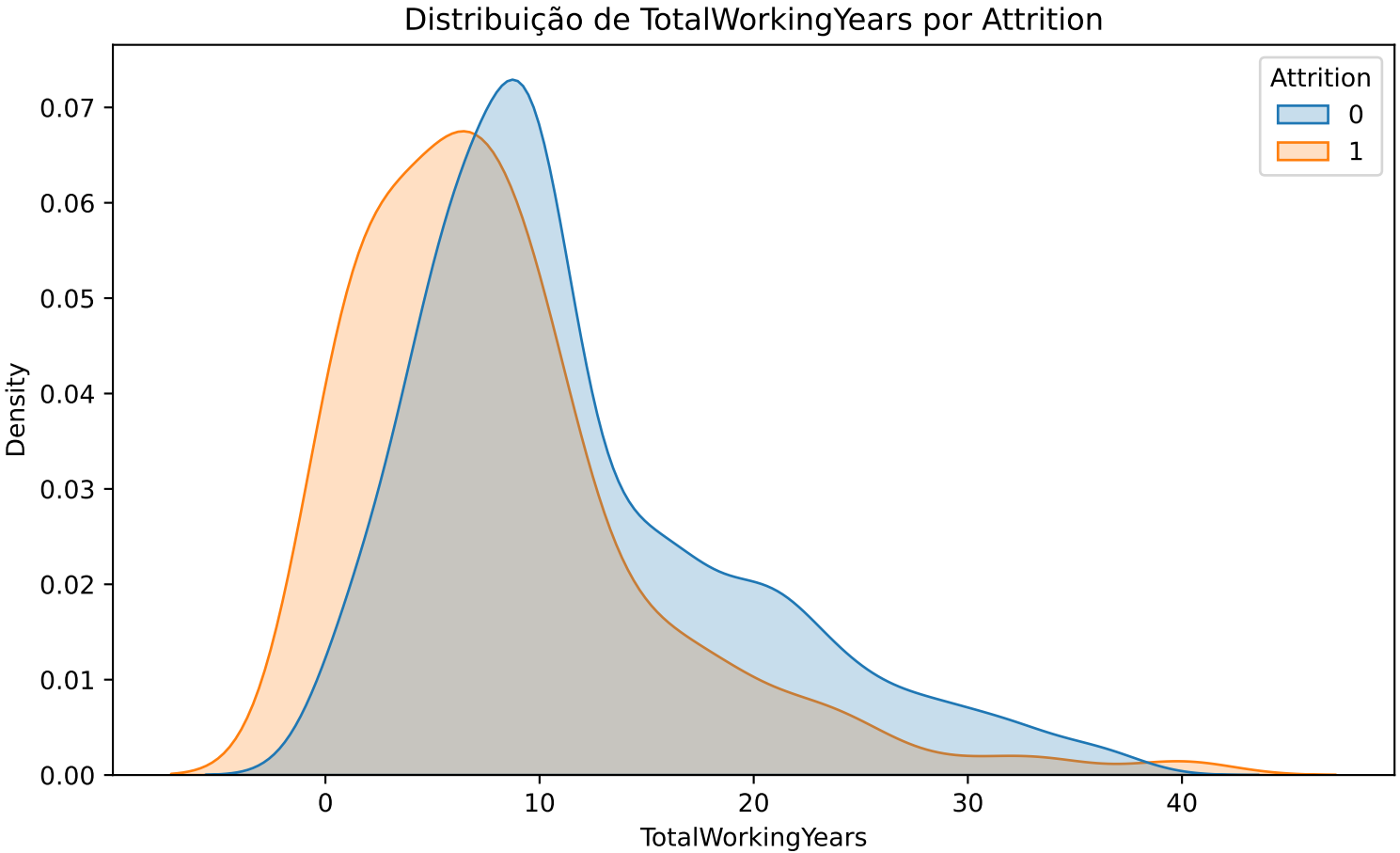
0
1
2
3
4
StockOptionLevel


Distribuição de TotalWorkingYears por Attrition
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Density
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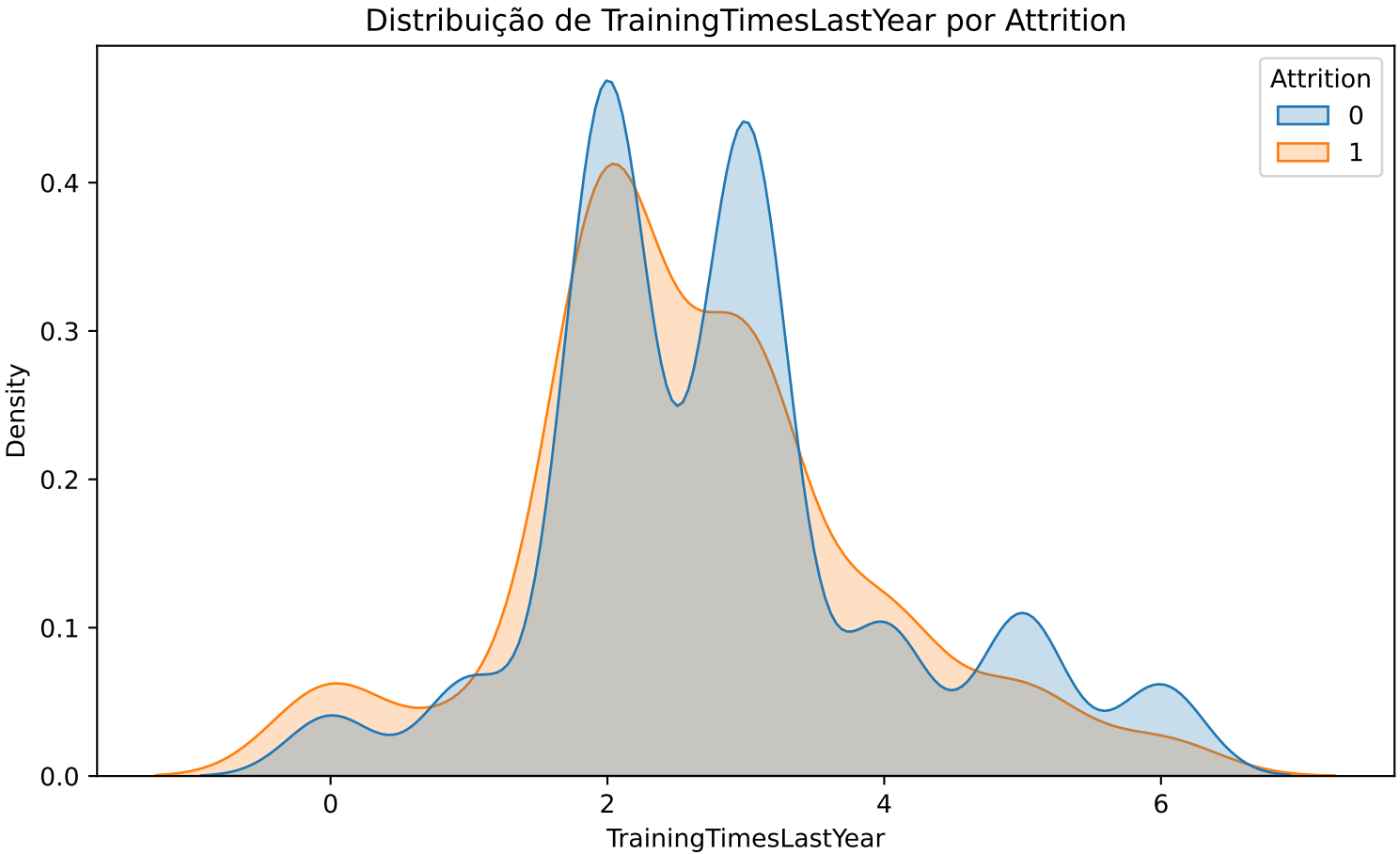
20
30
40
TotalWorkingYears


Distribuição de TrainingTimesLastYear por Attrition
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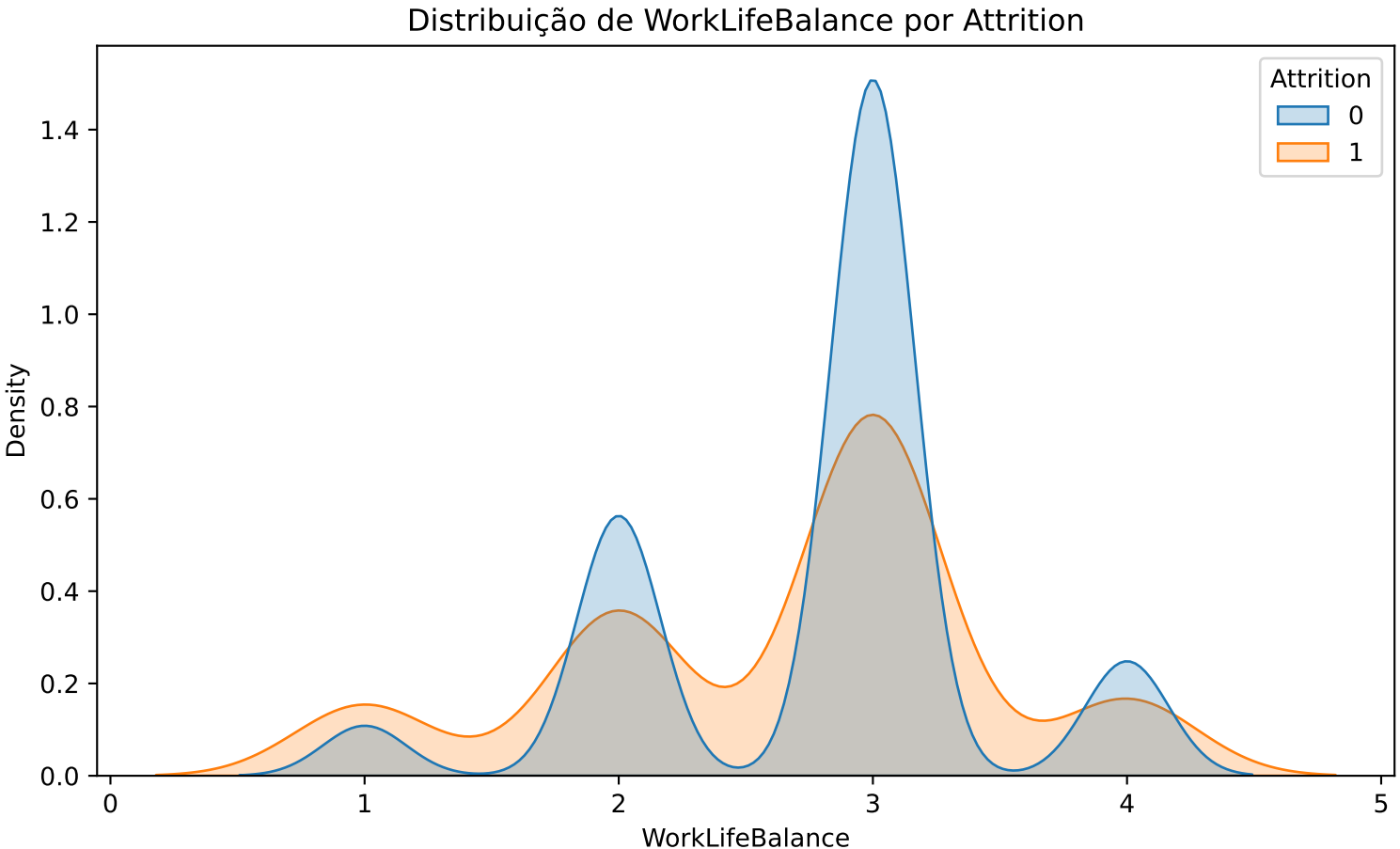
2
4
6
TrainingTimesLastYear


Distribuição de WorkLifeBalance por Attrition
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1.2
1.4
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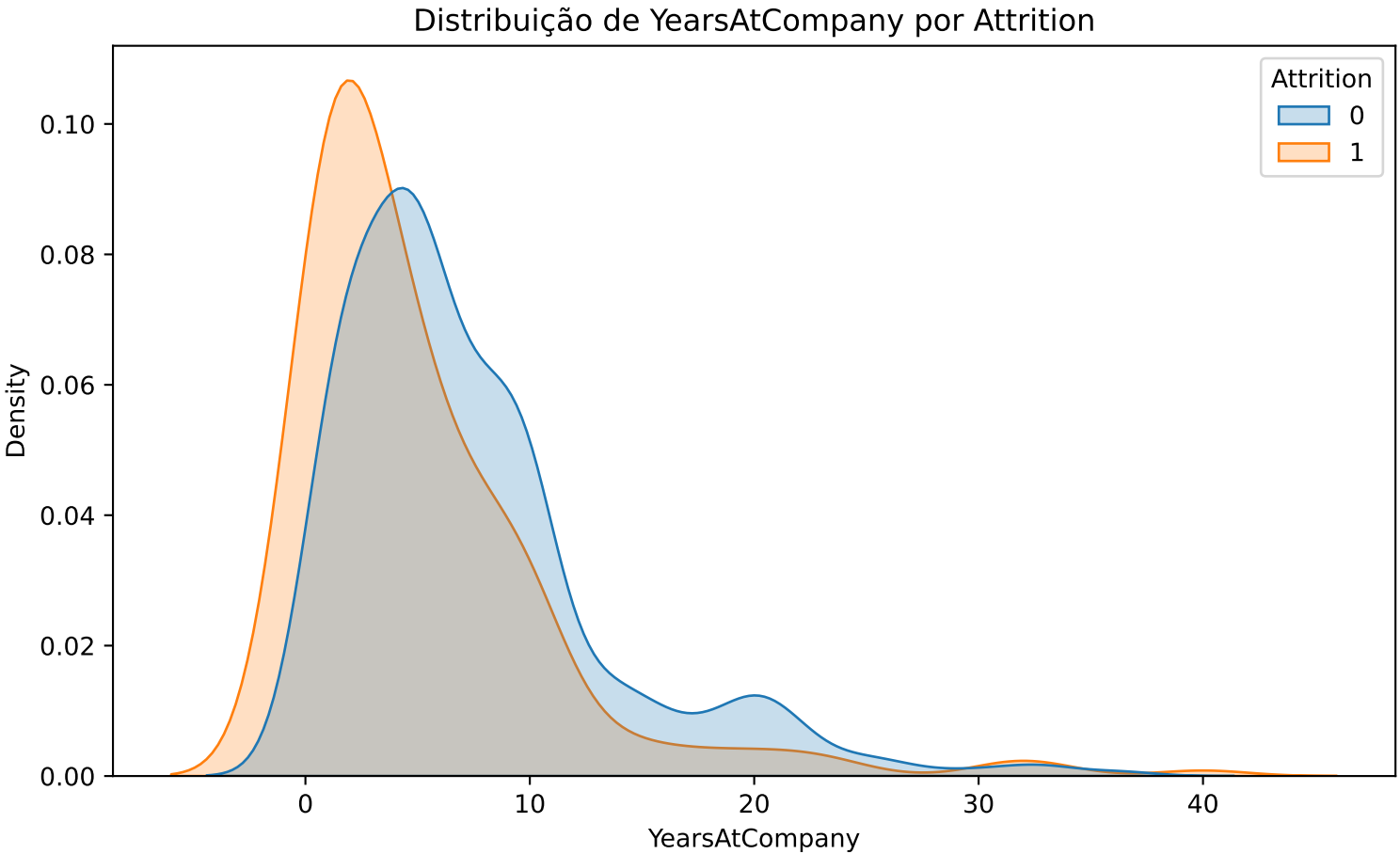
1
2
3
4
5
WorkLifeBalance


Distribuição de YearsAtCompany por Attrition
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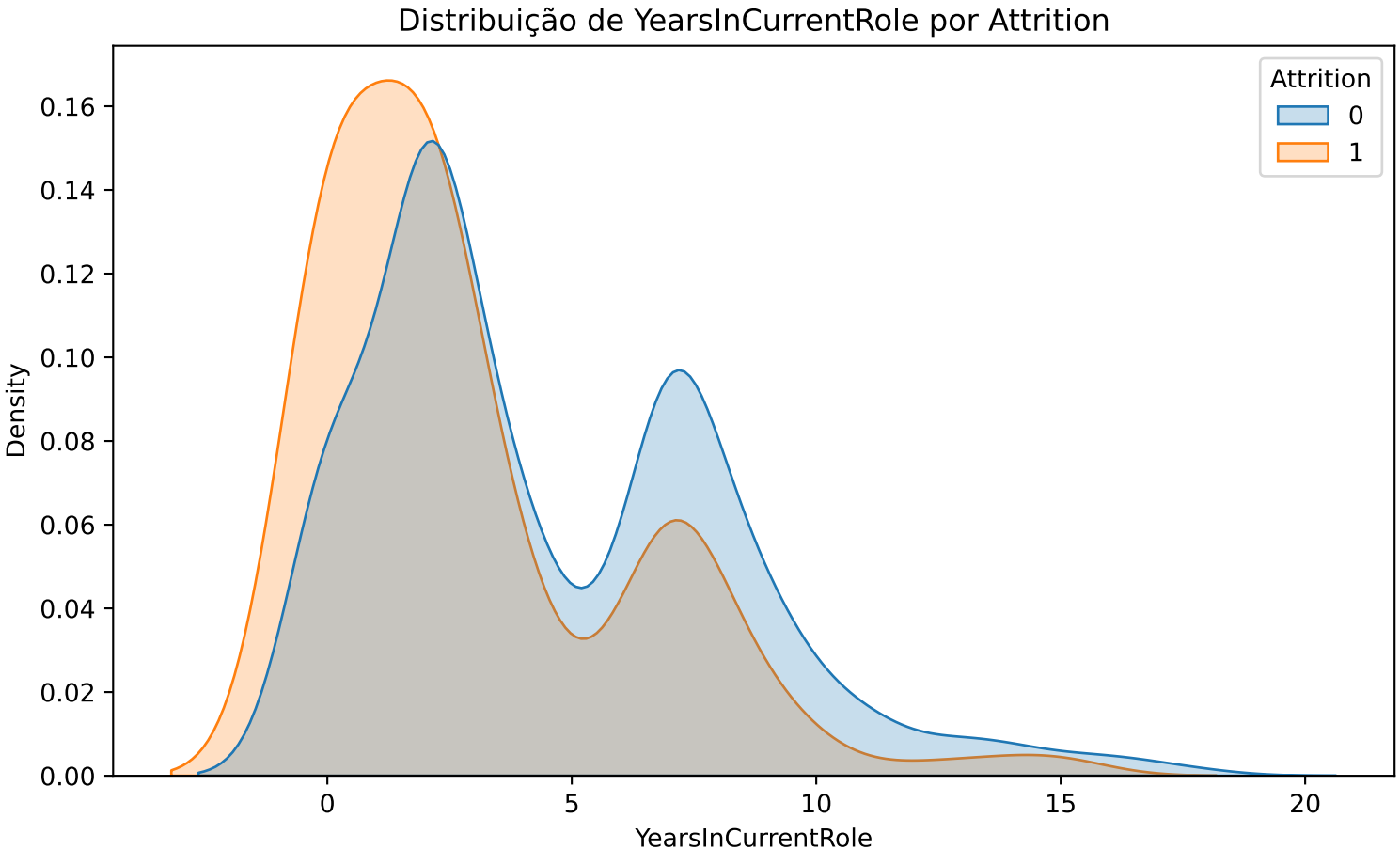
YearsAtCompany


Distribuição de YearsInCurrentRole por Attrition
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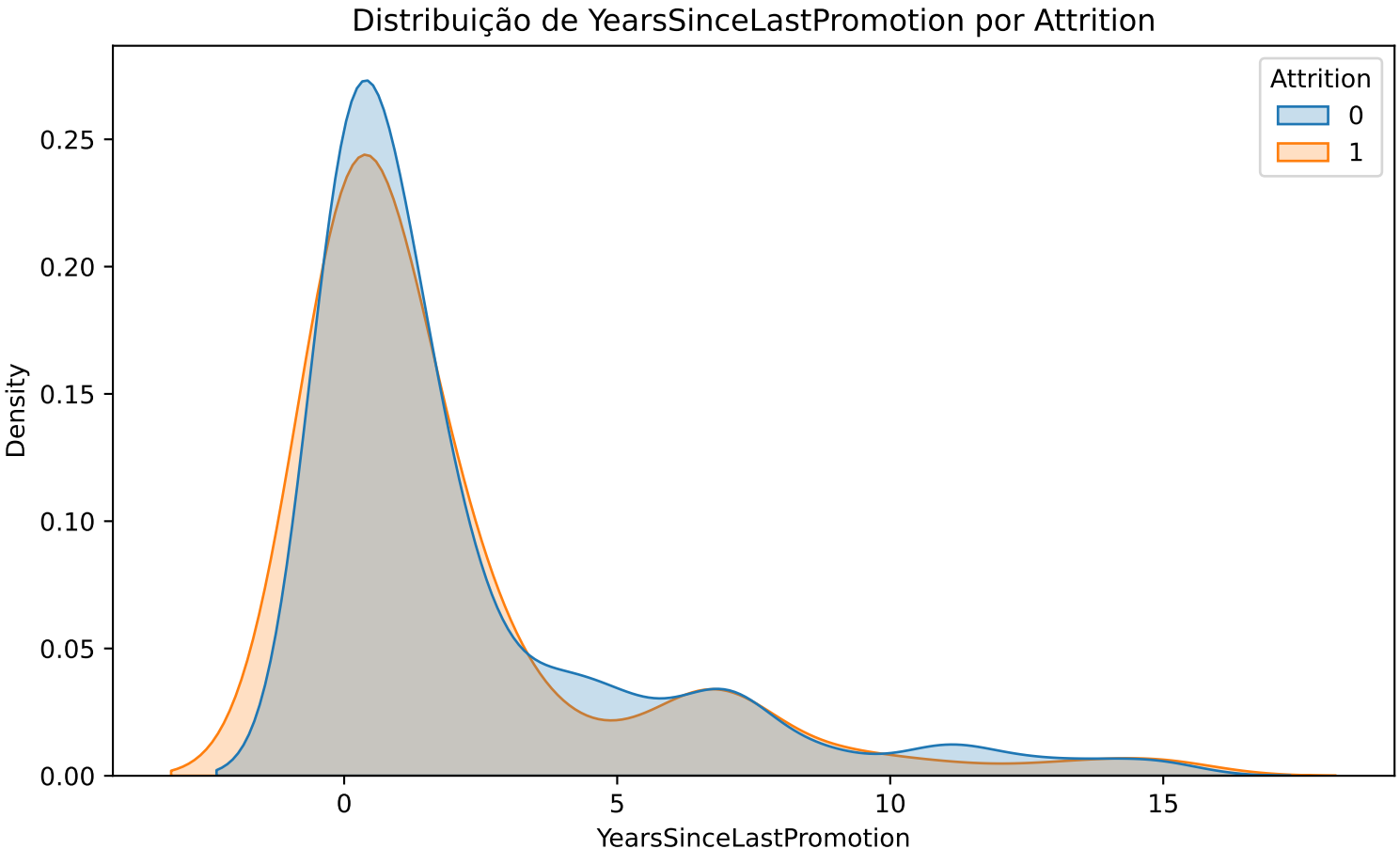
20
YearsInCurrentRole


Distribuição de YearsSinceLastPromotion por Attrition
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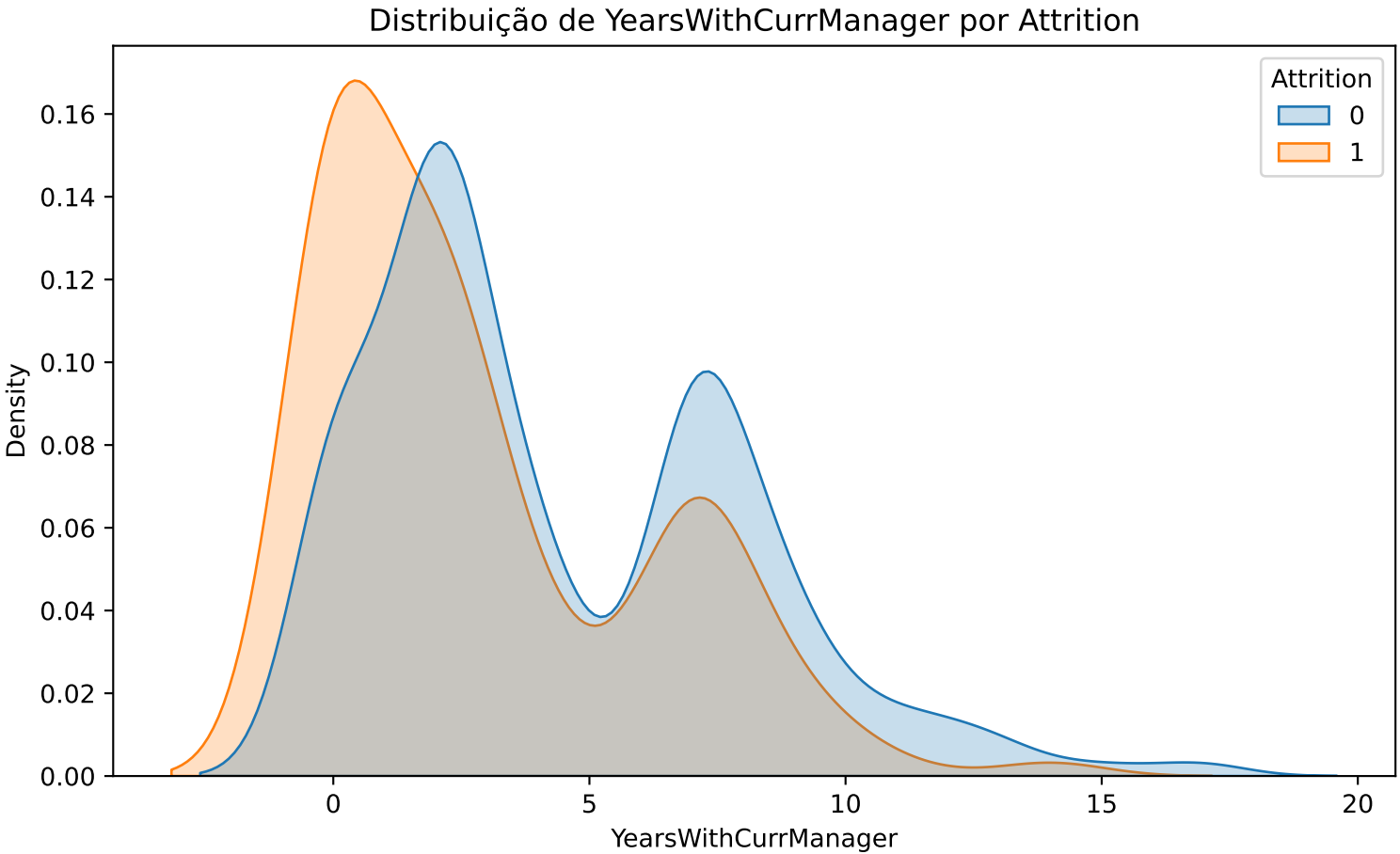
5
10
15
YearsSinceLastPromotion


Distribuição de YearsWithCurrManager por Attrition
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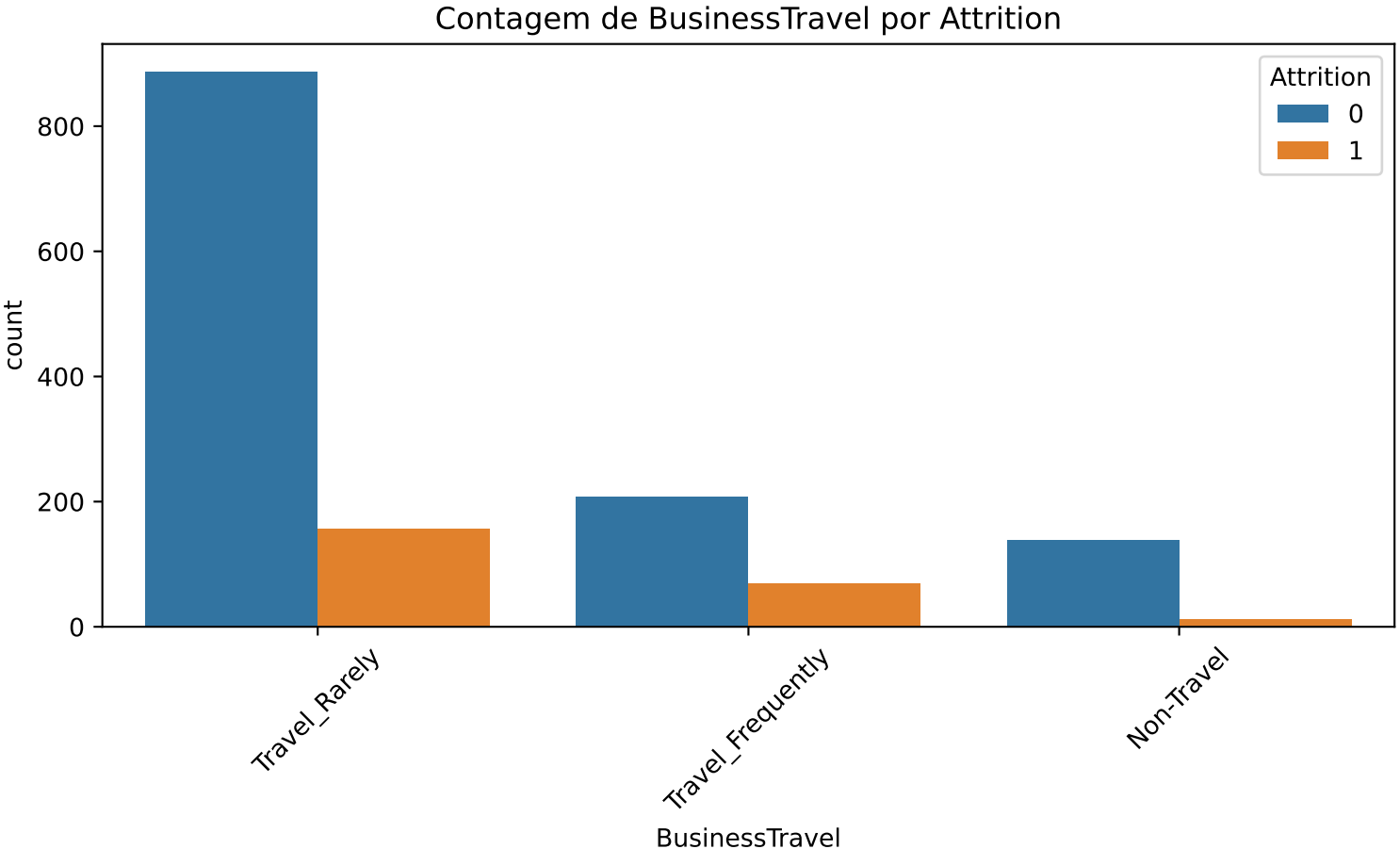
5
10
15
20
YearsWithCurrManager


Contagem de BusinessTravel por Attrition
Attrition
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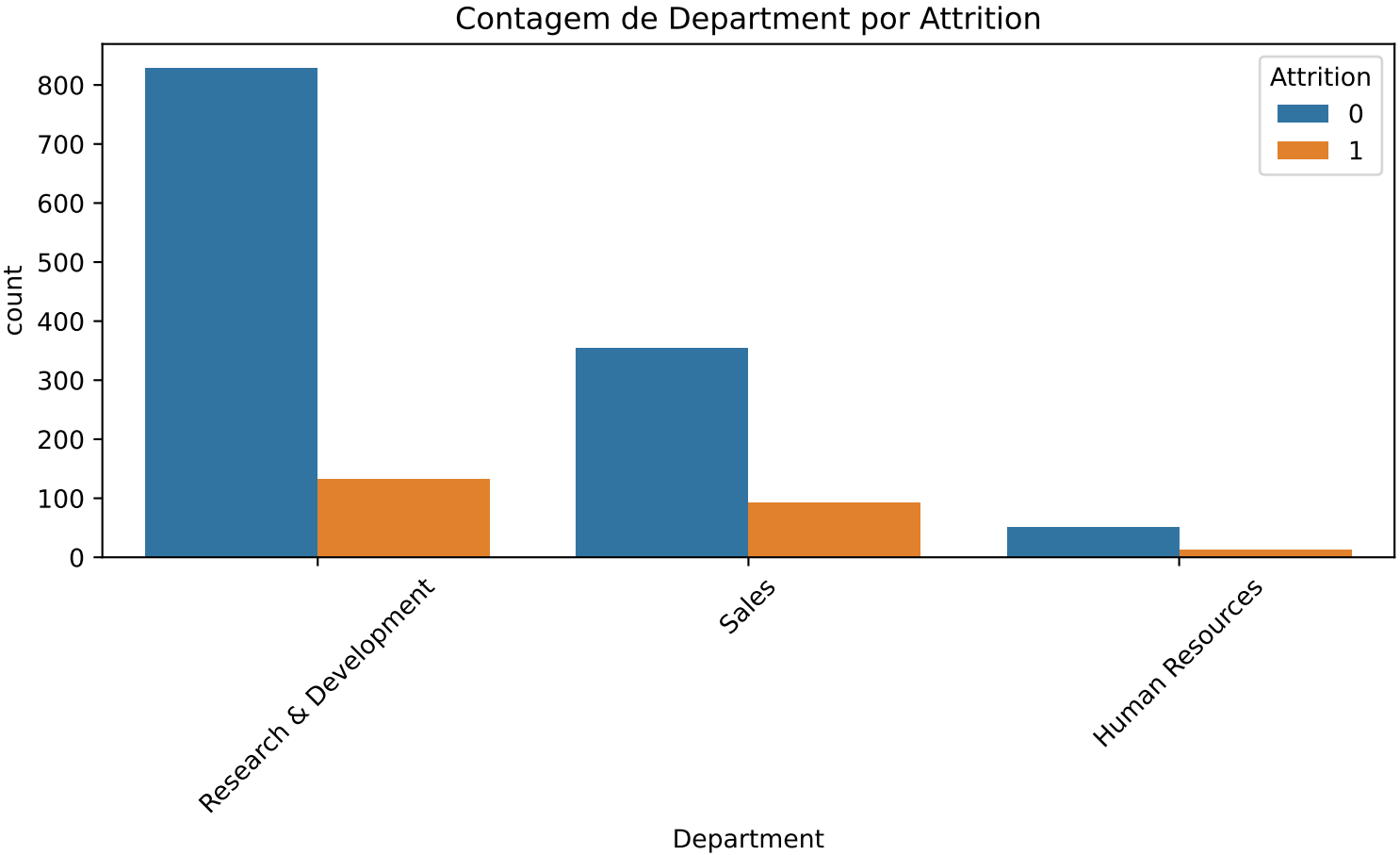
1
count
0
200
400
600
800
Travel_Rarely
Travel_Frequently
Non-Travel
BusinessTravel


Contagem de Department por Attrition
Attrition
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1
count
800
700
600
500
400
300
200
100
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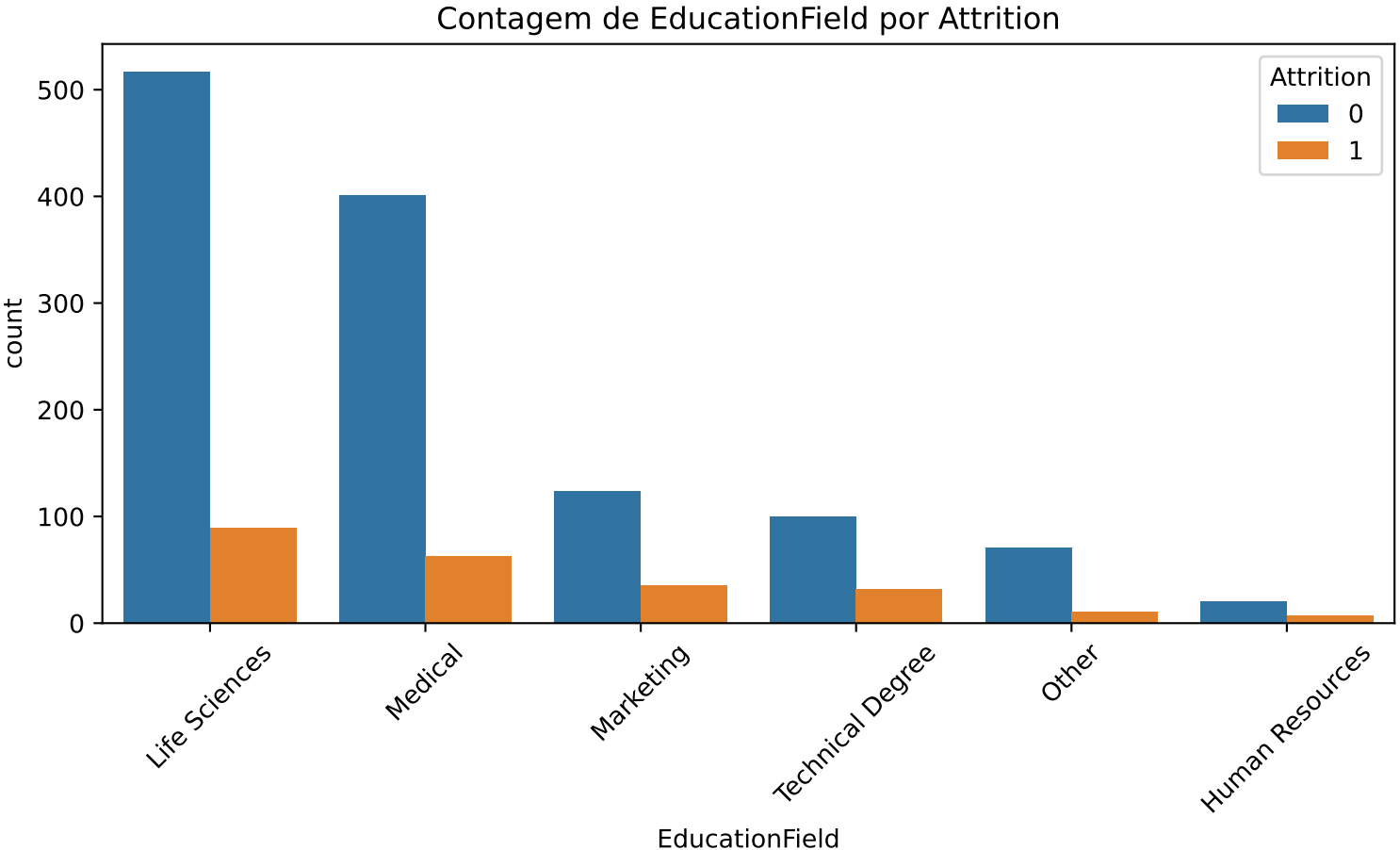
Research & Development
Sales
Human Resources
Department


Contagem de EducationField por Attrition
Attrition
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count
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400
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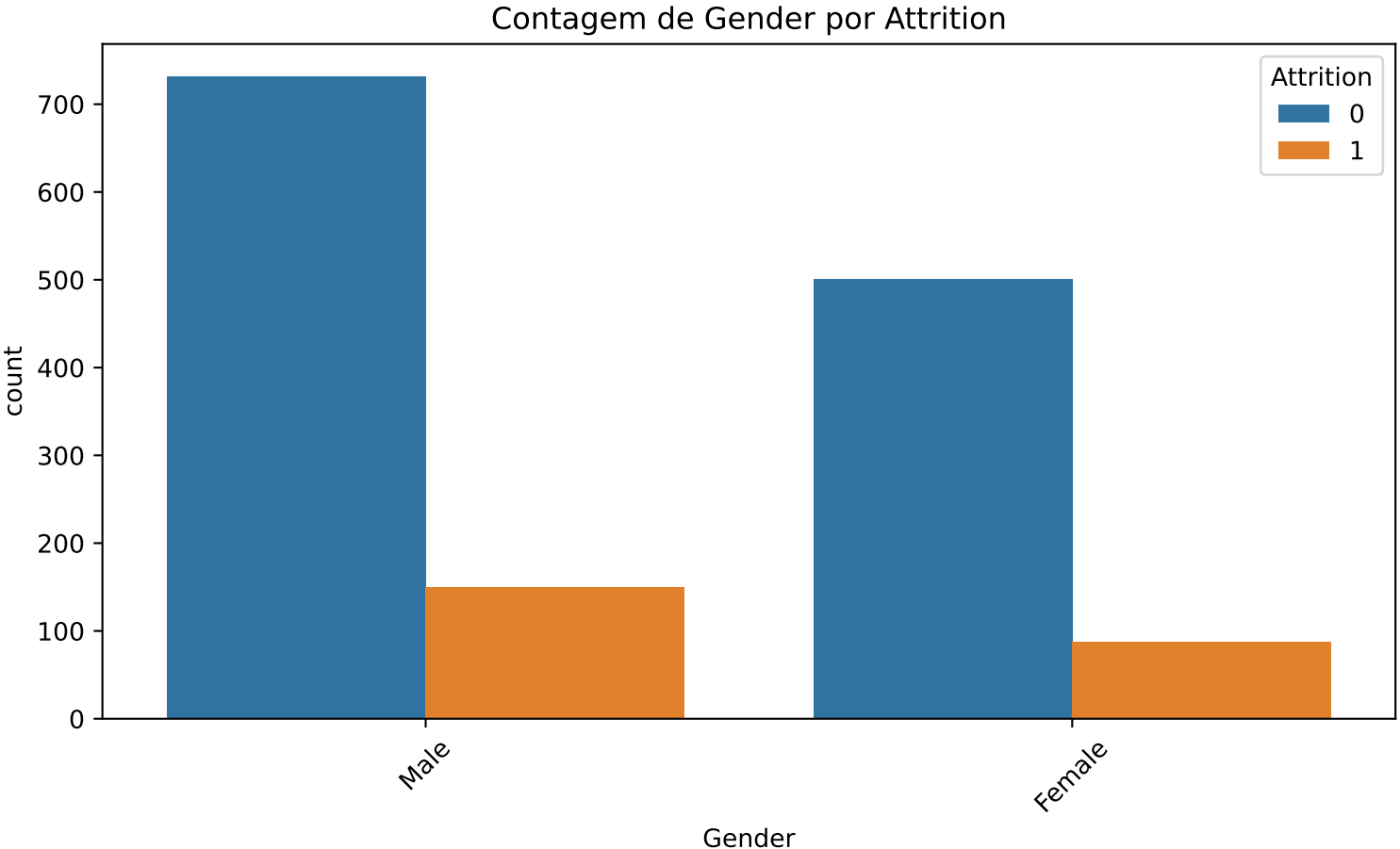
Life Sciences
Medical
Marketing
Technical Degree
Other
Human Resources
EducationField


Contagem de Gender por Attrition
Attrition
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1
count
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200
300
400
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600
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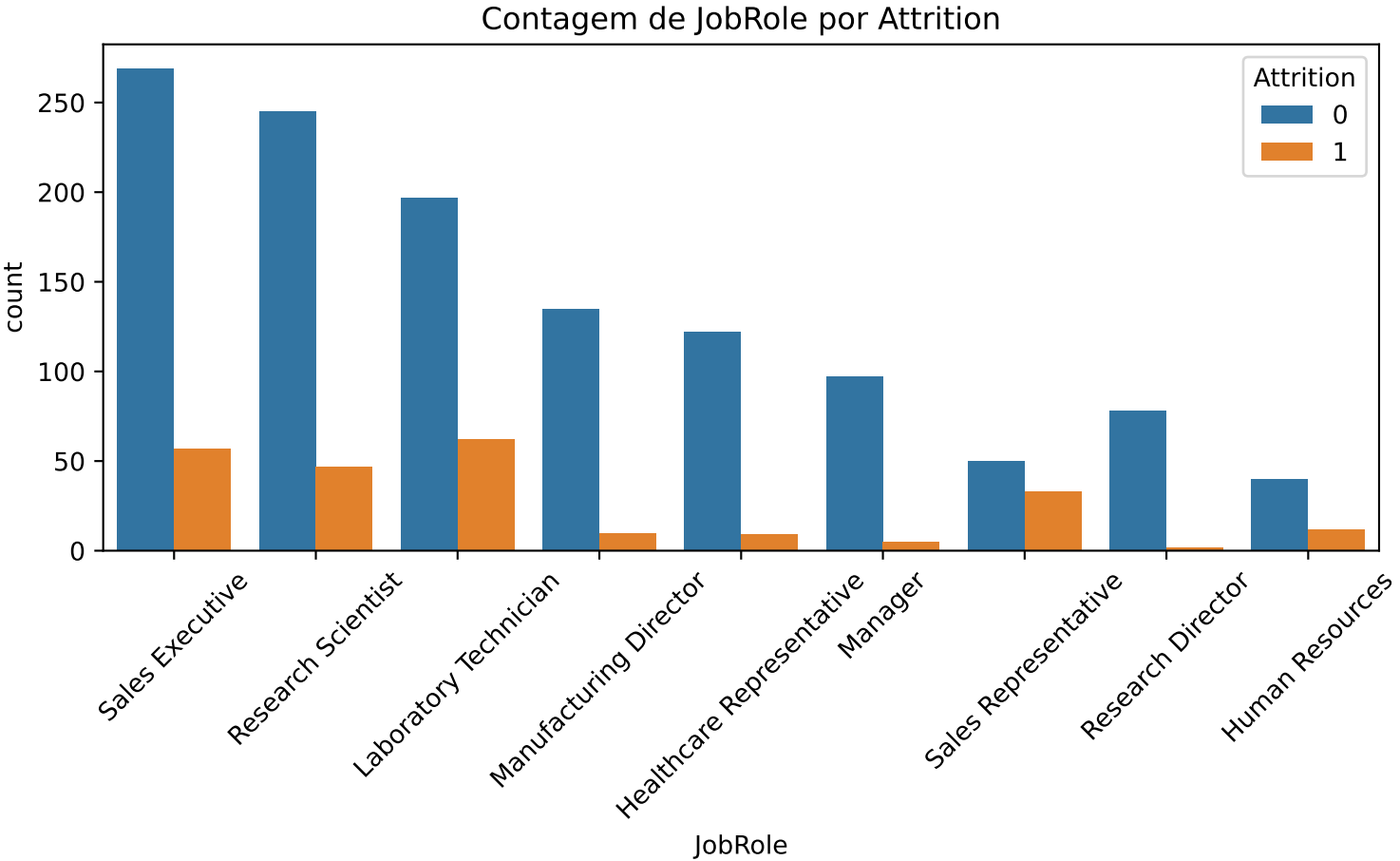
Male
Female
Gender


Contagem de JobRole por Attrition
Attrition
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1
count
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100
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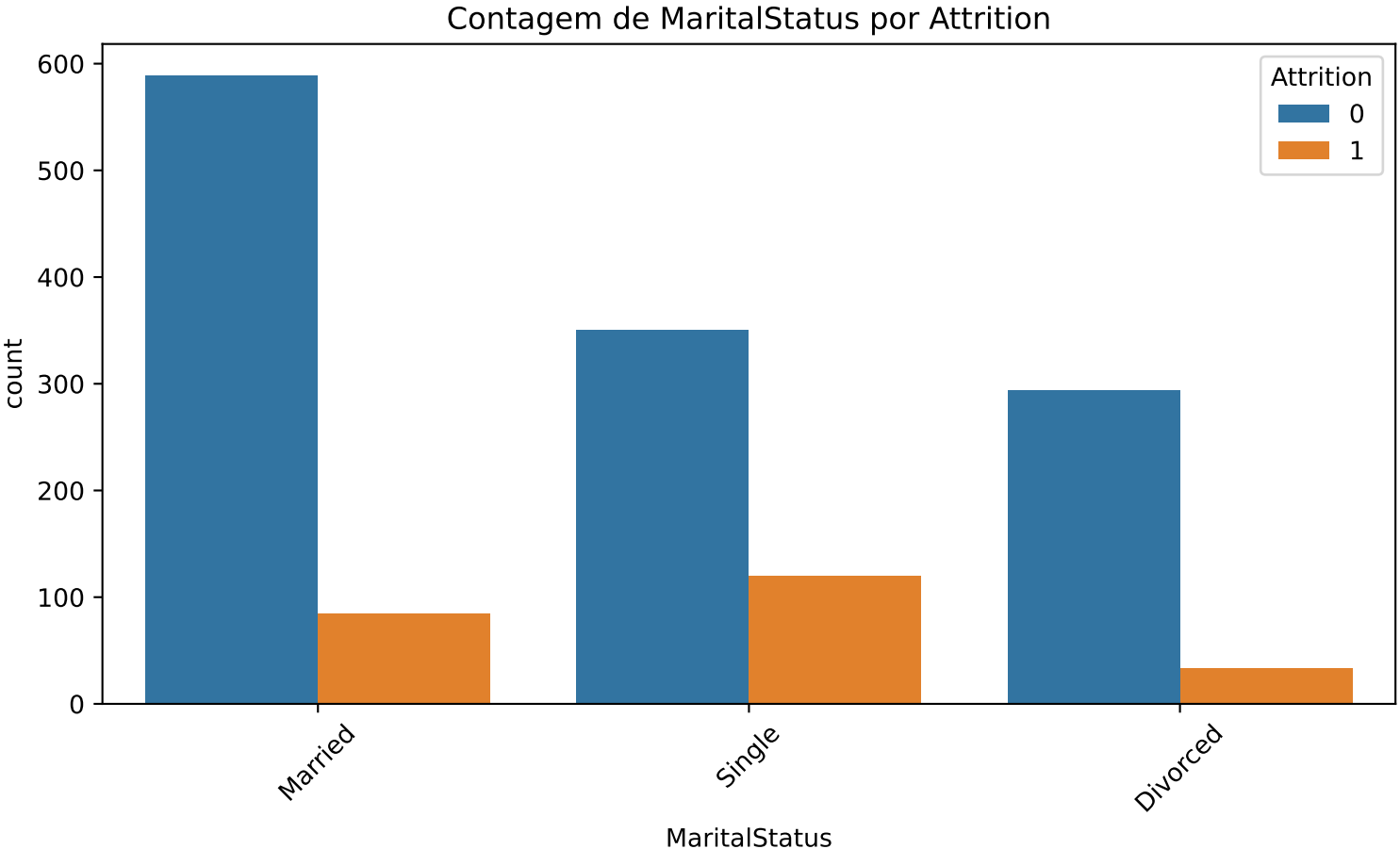
Sales Executive
Research Scientist
Laboratory Technician
Manufacturing Director
Healthcare Representative
Manager
Sales Representative
Research Director
Human Resources
JobRole


Contagem de MaritalStatus por Attrition
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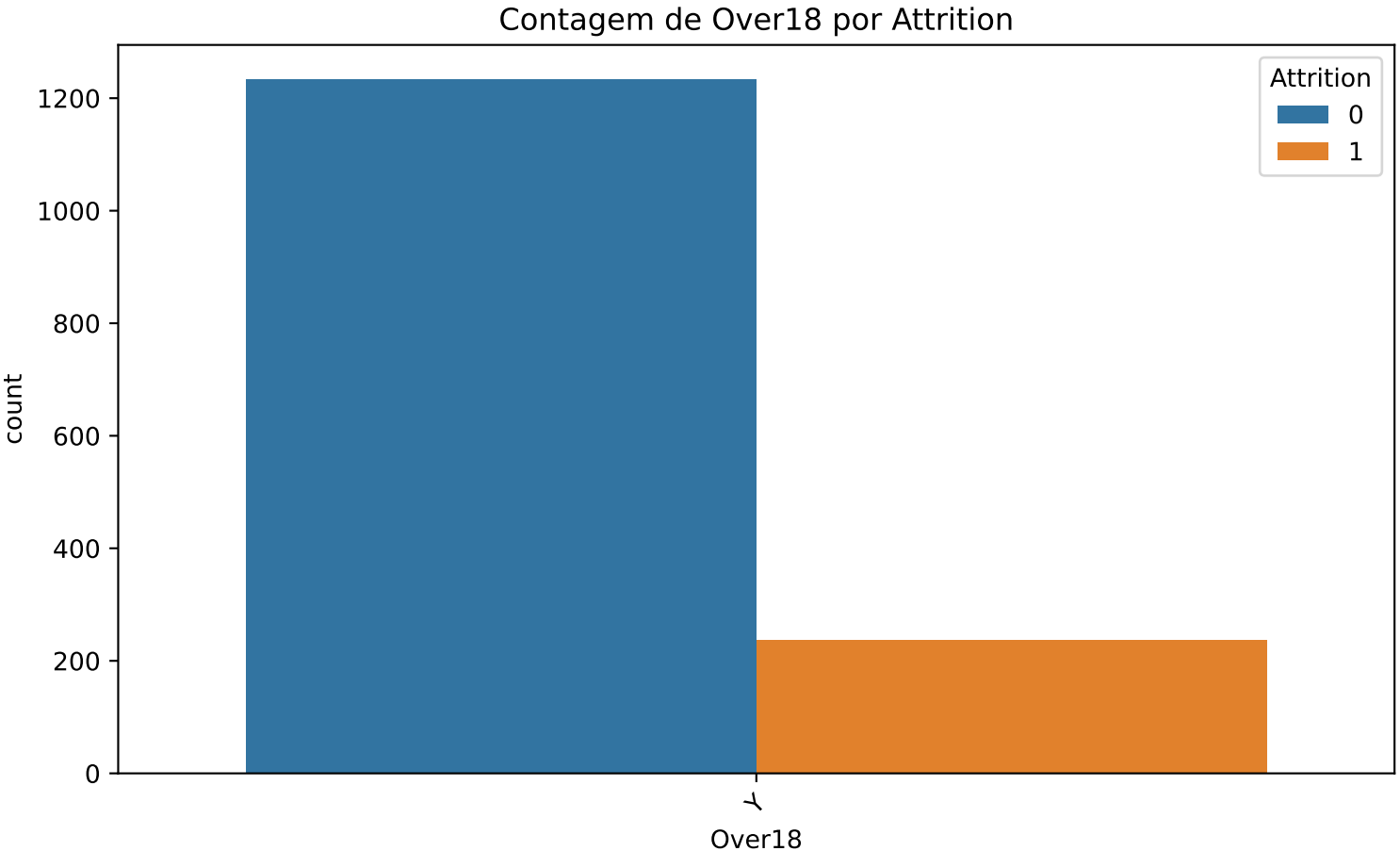
Married
Single
Divorced
MaritalStatus


Contagem de Over18 por Attrition
Attrition
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count
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1000
800
600
400
200
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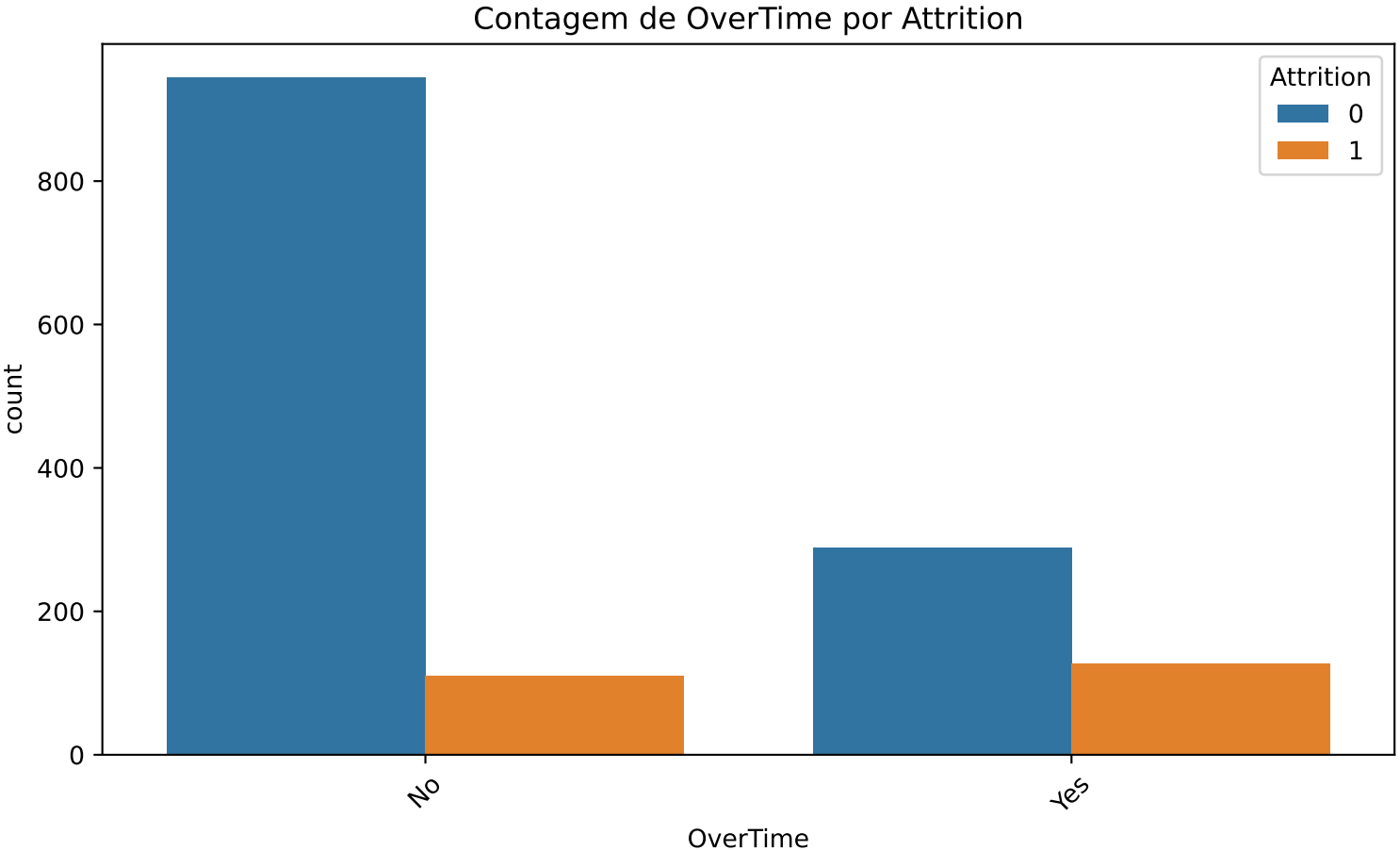
Over18


Contagem de OverTime por Attrition
Attrition
0
1
count
800
600
400
200
0
No
Yes
OverTime

Leaderboard de Modelos (Cross-Validation)

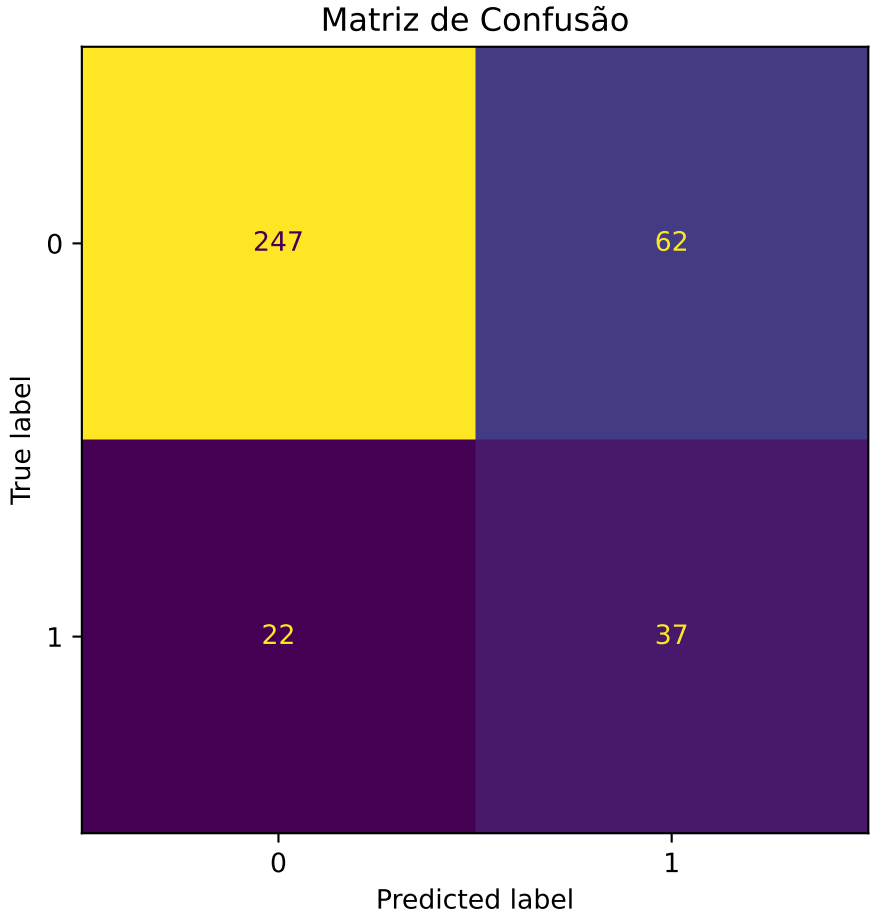
model	f1_cv_mean	f1_cv_std	Melhor?
logreg	0.5282974478405661	0.040235787539605966	▯
gb	0.4109260307896744	0.011677534645676925	
rf	0.23022172949002218	0.056176189166595086	

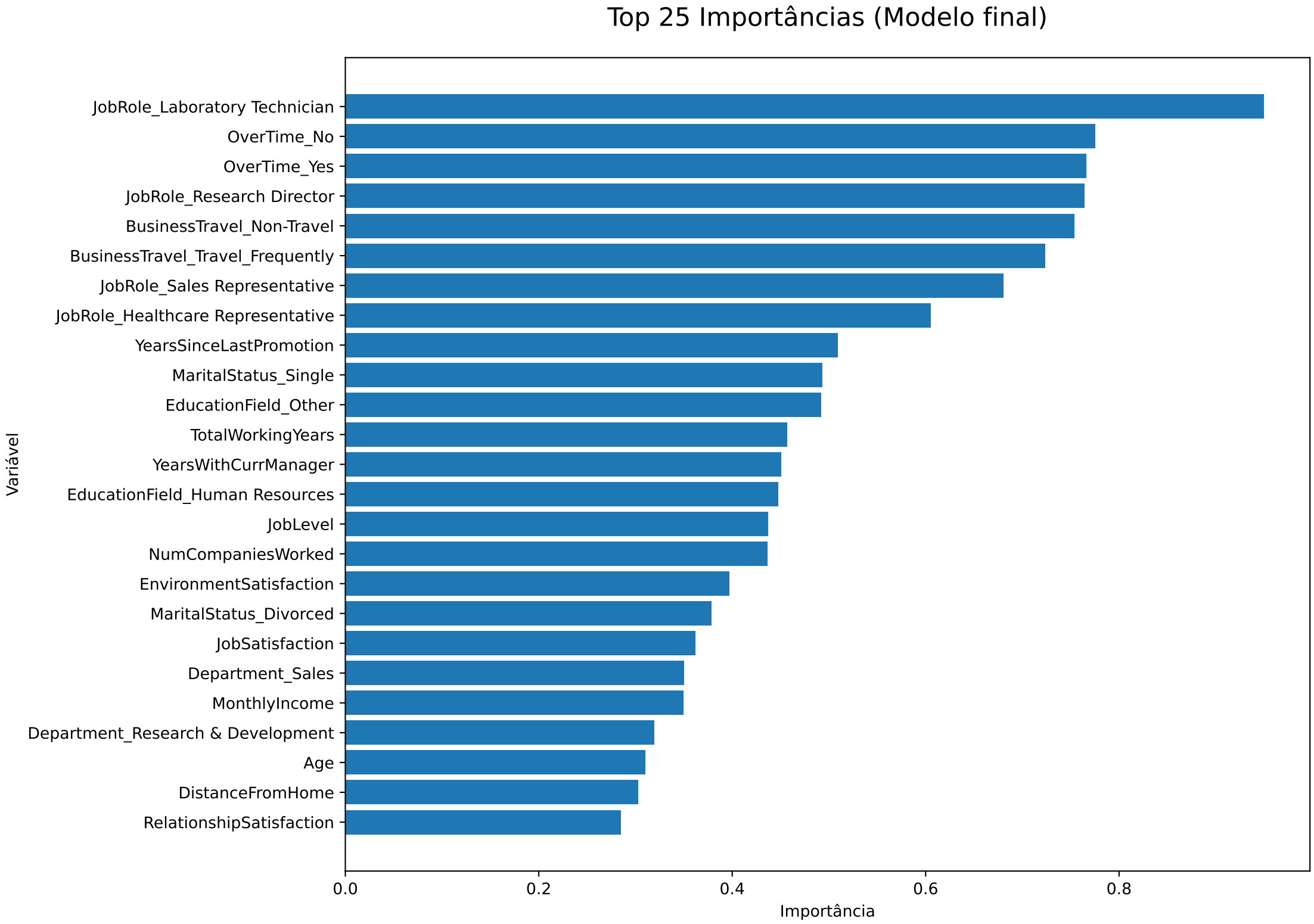
Critério de seleção: F1 médio no CV (maior é melhor).

Avaliação no Hold-out (Teste)

Modelo final (tunado): logreg

F1: 0.4684
Precision: 0.3737
Recall: 0.6271




Top 25 Importâncias (Modelo final)
JobRole_Laboratory Technician
OverTime_No
OverTime_Yes
JobRole_Research Director
BusinessTravel_Non-Travel
BusinessTravel_Travel_Frequently
JobRole_Sales Representative
JobRole_Healthcare Representative
YearsSinceLastPromotion
MaritalStatus_Single
EducationField_Other
TotalWorkingYears
YearsWithCurrManager
EducationField_Human Resources
JobLevel
NumCompaniesWorked
EnvironmentSatisfaction
MaritalStatus_Divorced
JobSatisfaction
Department_Sales
MonthlyIncome
Department_Research & Development
Age
DistanceFromHome
RelationshipSatisfaction
Variável
0.0
0.2
0.4
0.6
0.8
Importância